

Biología para informáticos

Máster en Sistemas Inteligentes

Luis Quintales
Departamento de Informática



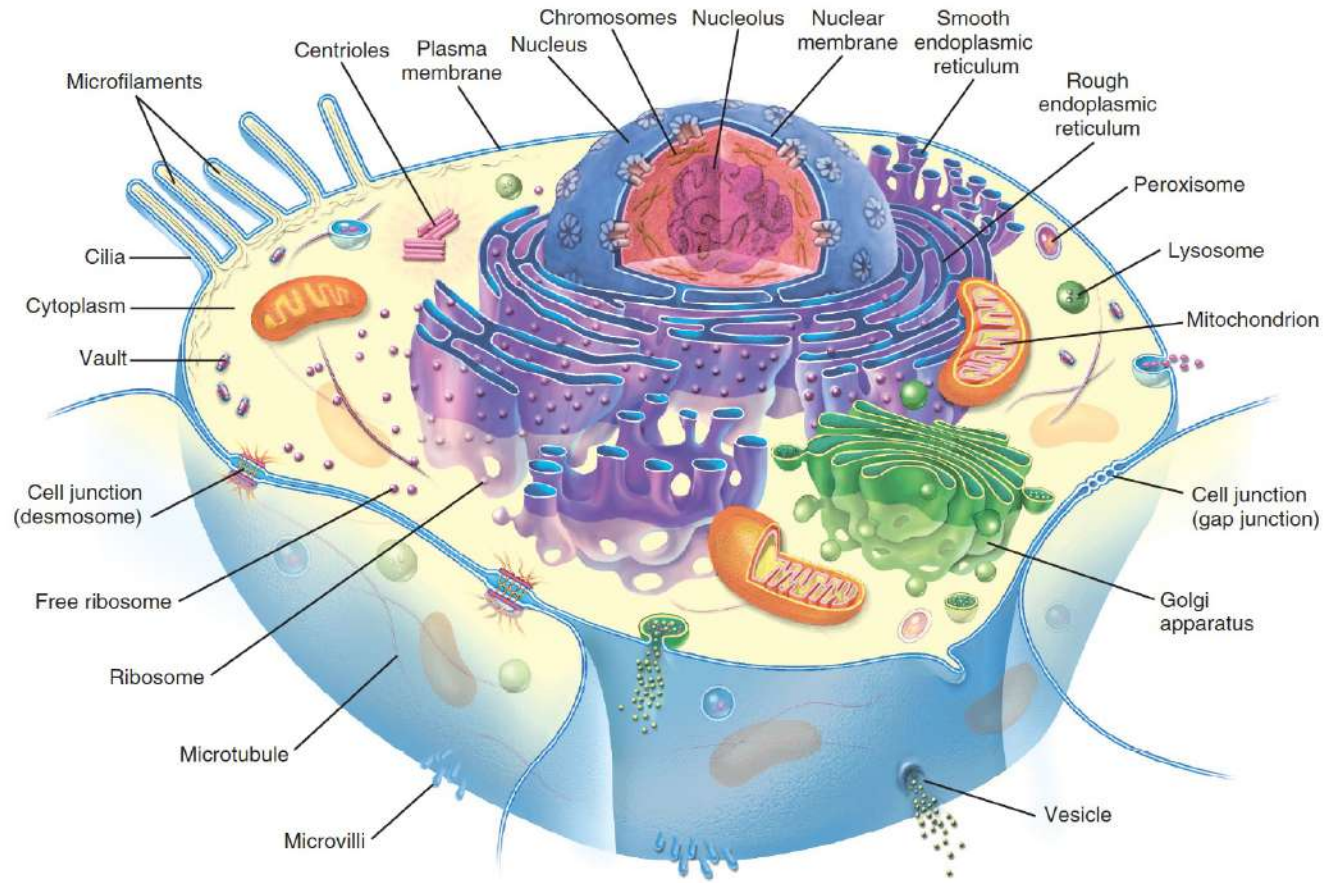
El reto: conocer el genoma de cada uno de nosotros



Índice

- Células, genomas y cromosomas
- DNA
- Replicación
- Genes, transcripción y proteínas
- Mutaciones
- Genotipo y fenotipo

Una célula eucariota



Número de células en un humano

Table 3. B/H ratio for different population. See Table B in [S1 Appendix](#) for full references.

population segment	body weight [kg]	age [y]	blood volume [L]	RBC count [$10^{12}/L$]	colon content [g]	bac. conc. [$10^{11}/g$ wet] ⁽¹⁾	total human cells [10^{12}] ⁽²⁾	total bacteria [10^{12}]	B:H
ref. man	70	20–30	4.9	5.0	420	0.92	30	38	1.3
ref. woman	63		3.9	4.5	480	0.92	21	44	2.2
young infant	4.4	4 weeks	0.4	3.8	48	0.92	1.9	4.4	2.3
infant	9.6	1	0.8	4.5	80	0.92	4	7	1.7
elder	70	66	3.8 ⁽³⁾	4.8	420	0.92	22	38	1.8
obese	140		6.7	5.0 ⁽⁴⁾	610 ⁽⁵⁾	0.92	40	56	1.4

ESSAY

Revised Estimates for the Number of Human and Bacteria Cells in the Body

Ron Sender¹, Shai Fuchs^{2*}, Ron Milo^{1*}

¹ Department of Plant and Environmental Sciences, Weizmann Institute of Science, Rehovot, Israel,
² Department of Molecular Genetics, Weizmann Institute of Science, Rehovot, Israel

* Current address: Division of Endocrinology, The Hospital for Sick Children and Department of Pediatrics, The University of Toronto, Toronto, Canada

* shai.fuchs@sickkids.ca (SF); ron.milo@weizmann.ac.il (RM)



OPEN ACCESS

Citation: Sender R, Fuchs S, Milo R (2016) Revised Estimates for the Number of Human and Bacteria Cells in the Body. *PLoS Biol* 14(8): e1002533. doi:10.1371/journal.pbio.1002533

Published: August 19, 2016

Abstract

Reported values in the literature on the number of cells in the body differ by orders of magnitude and are very seldom supported by any measurements or calculations. Here, we integrate the most up-to-date information on the number of human and bacterial cells in the body. We estimate the total number of bacteria in the 70 kg "reference man" to be $3.8 \cdot 10^{13}$. For human cells, we identify the dominant role of the hematopoietic lineage to the total count ($\approx 90\%$) and revise past estimates to $3.0 \cdot 10^{13}$ human cells. Our analysis also updates the widely-cited 10:1 ratio, showing that the number of bacteria in the body is actually of the same order as the number of human cells, and their total mass is about 0.2 kg.

Sender R, Fuchs S, Milo R.

Revised Estimates for the Number of Human and Bacteria Cells in the Body.

PLoS Biol. 2016;14(8):e1002533. Published 2016 Aug 19.

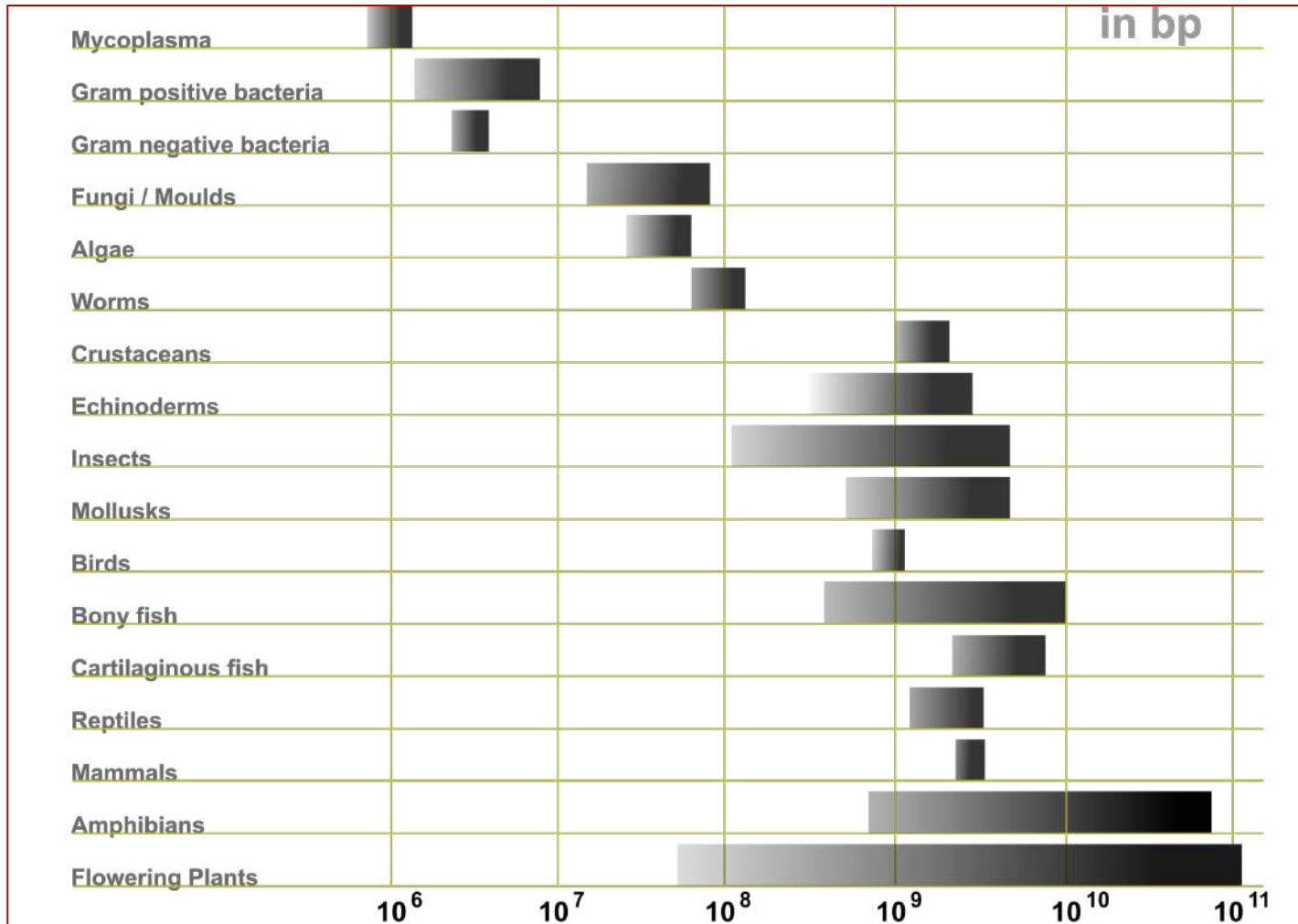
doi:10.1371/journal.pbio.1002533

Dimensiones del genoma humano

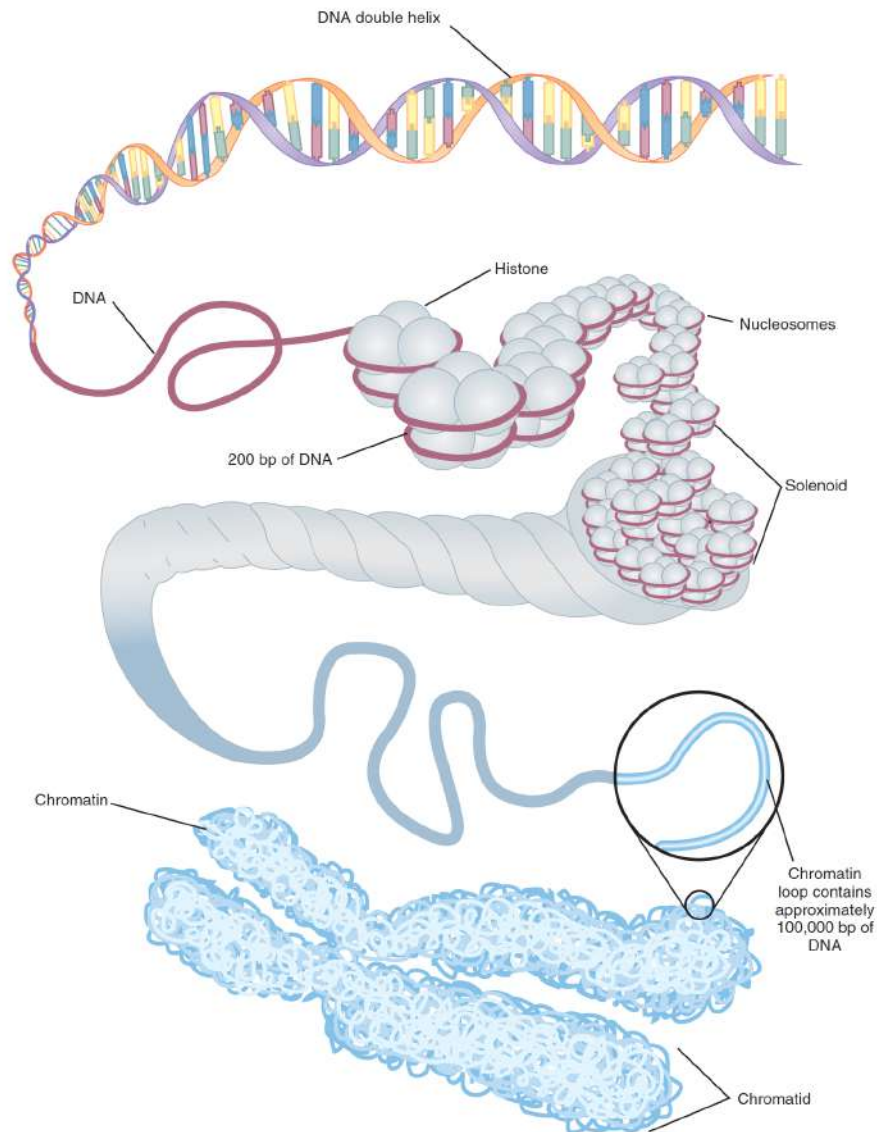


Riccardo Sabatini
How to read the genome and build a human being
[TED, 2016](#)

El tamaño de un genoma (I)



Empaquetamiento del DNA

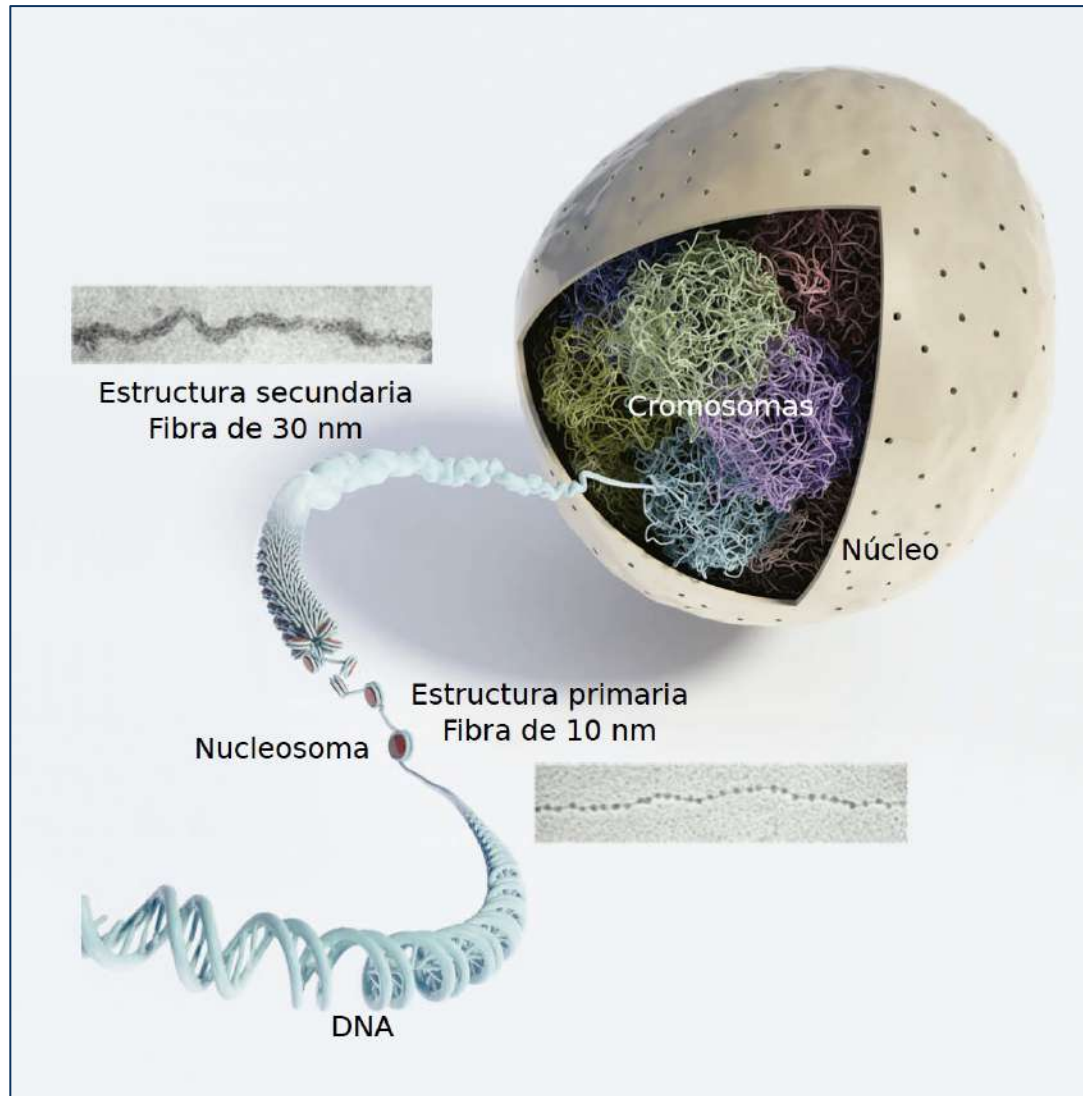


Empaquetamiento del DNA



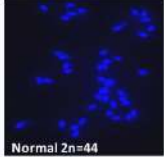
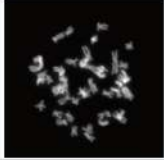
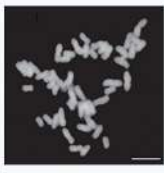
¿Cómo se pueden confinar 2 metros de DNA
en un núcleo de 6 micras?

Niveles de empaquetamiento de la cromatina



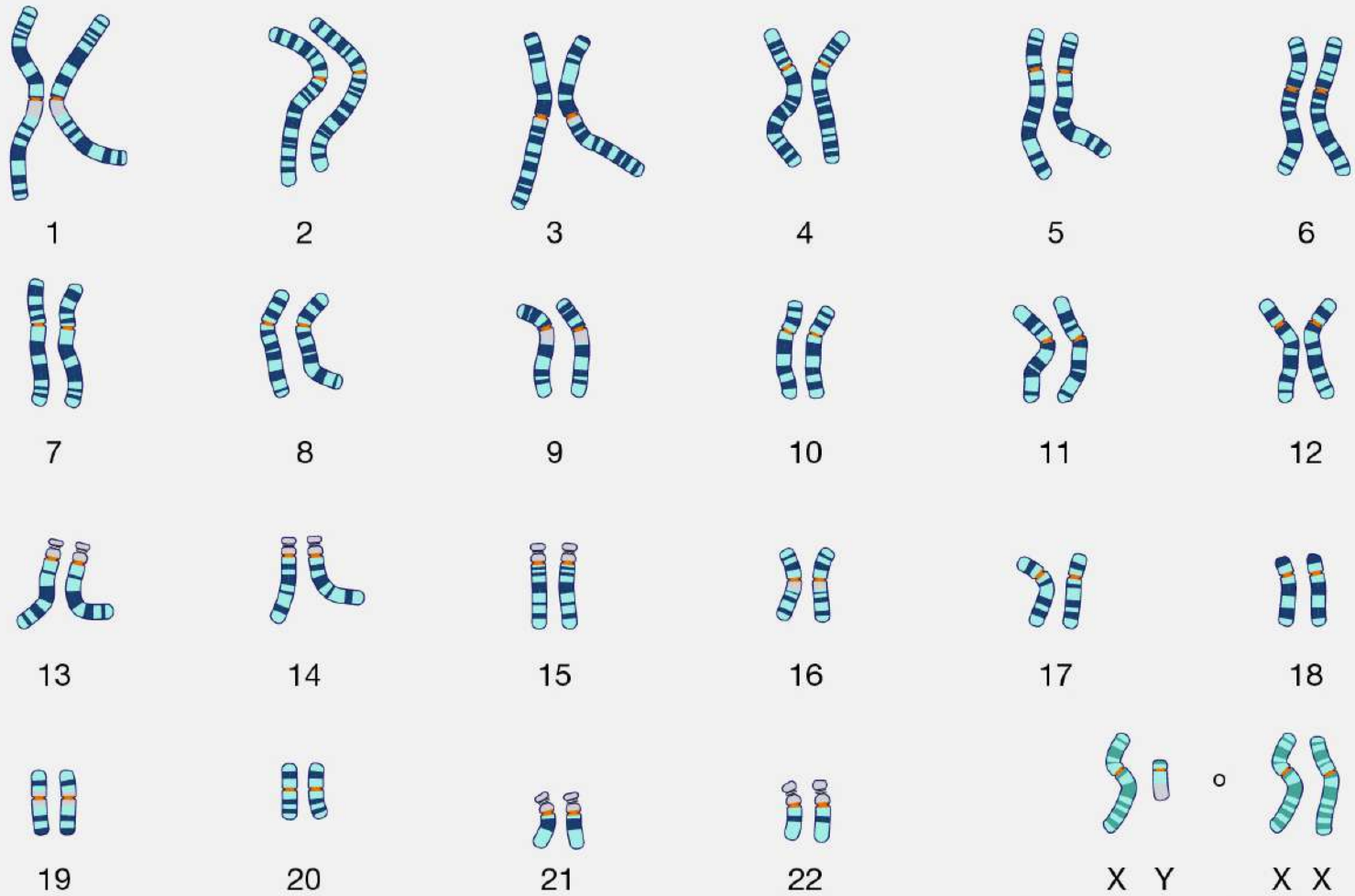
Adaptado de: Misteli T, *Scientific American* 2011 **304**:66-73

Cromosomas y organismos

Moon jellyfish (<i>Aurelia aurita</i>)	44		
Dolphin (Delphinidae)	44		
Arabian coffee (<i>Coffea arabica</i>)	44		 Normal 2n=44
Reeves's muntjac (<i>Muntiacus reevesi</i>)	46		
Human (<i>Homo sapiens</i>)	46		
Nilgai (<i>Boselaphus tragocamelus</i>)	46		
Parhyale hawaiensis	46		
Water buffalo (river type) (<i>Bubalus bubalis</i>)	48		
Tobacco (<i>Nicotiana tabacum</i>)	48		

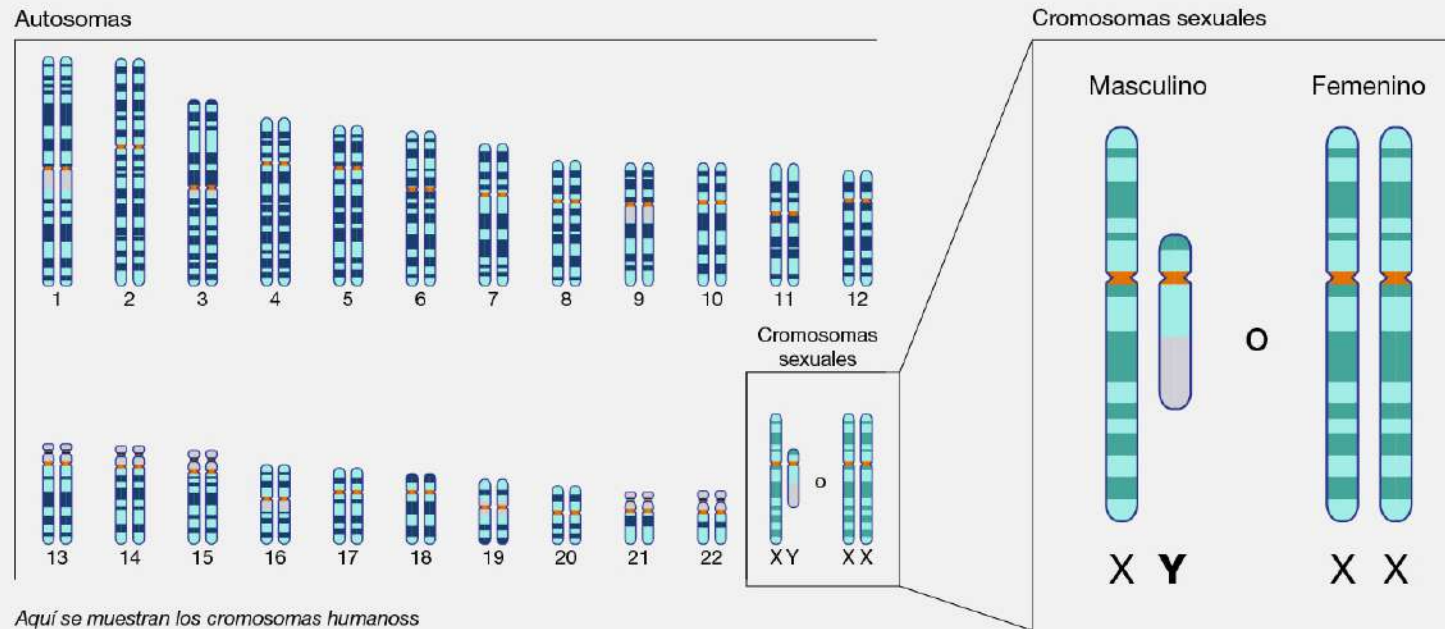
https://en.wikipedia.org/wiki/List_of_organisms_by_chromosome_count

Cariótipo

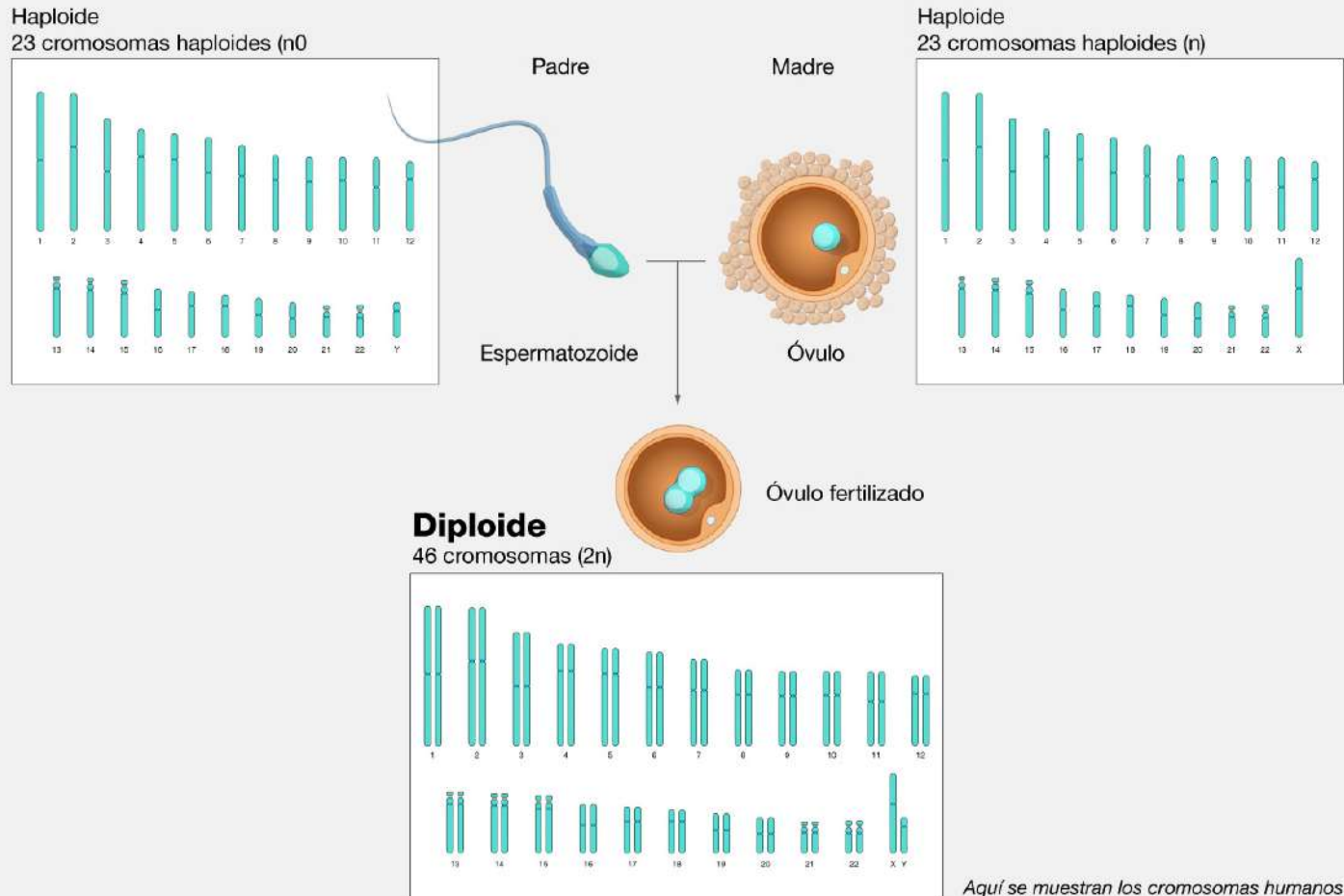


Aquí se muestran los cromosomas humanos

Autosomas y cromosomas sexuales



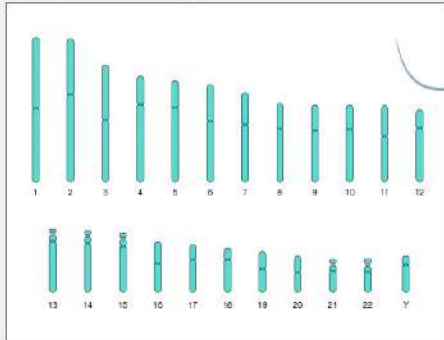
Diploide



Haploide

Haploide

23 cromosomas (n)



Padre

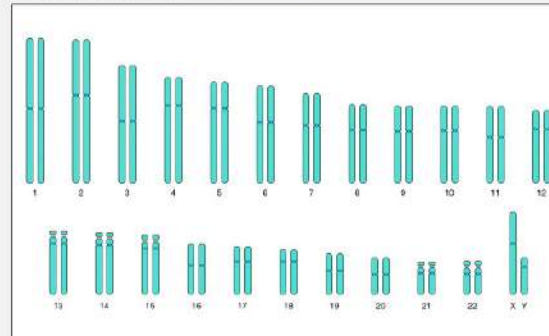
Madre

Espermatozoide

Óvulo

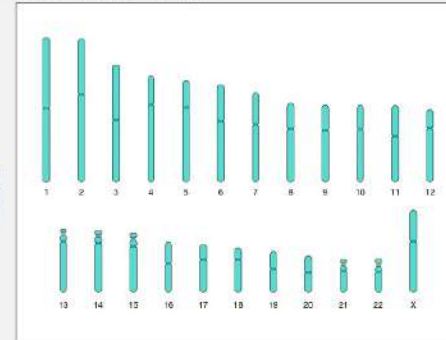
Óvulo fertilizado

Diploide
46 cromosomas (2n)



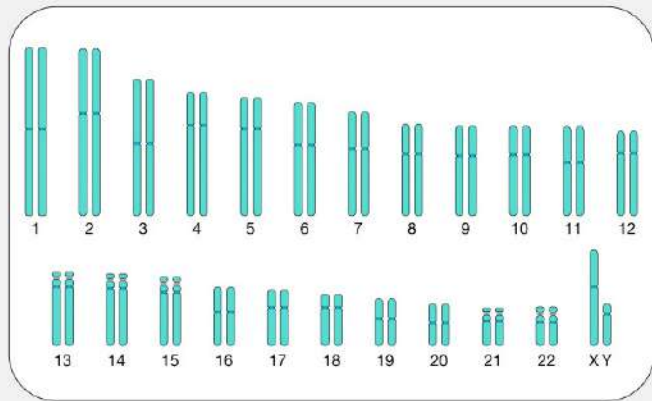
Haploide

23 cromosomas (n)

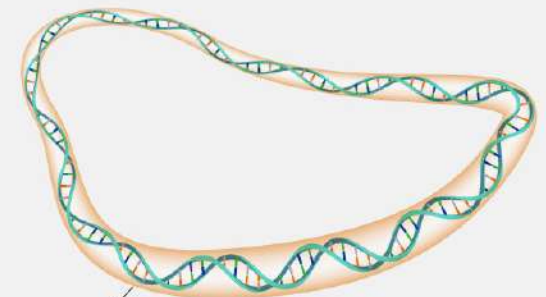
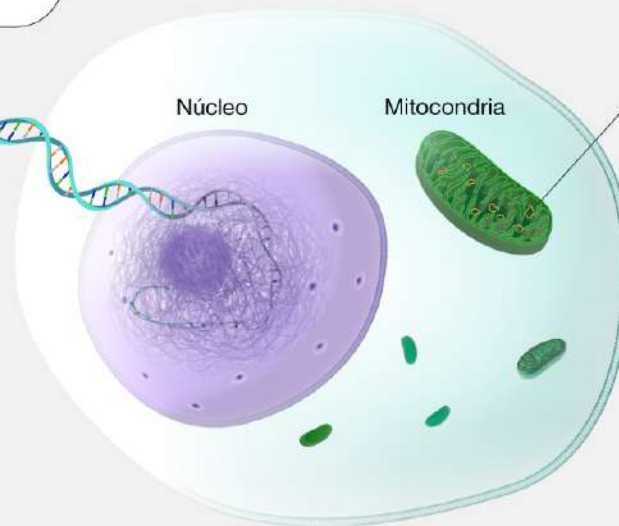


Aquí se muestran los cromosomas humanos

Genoma

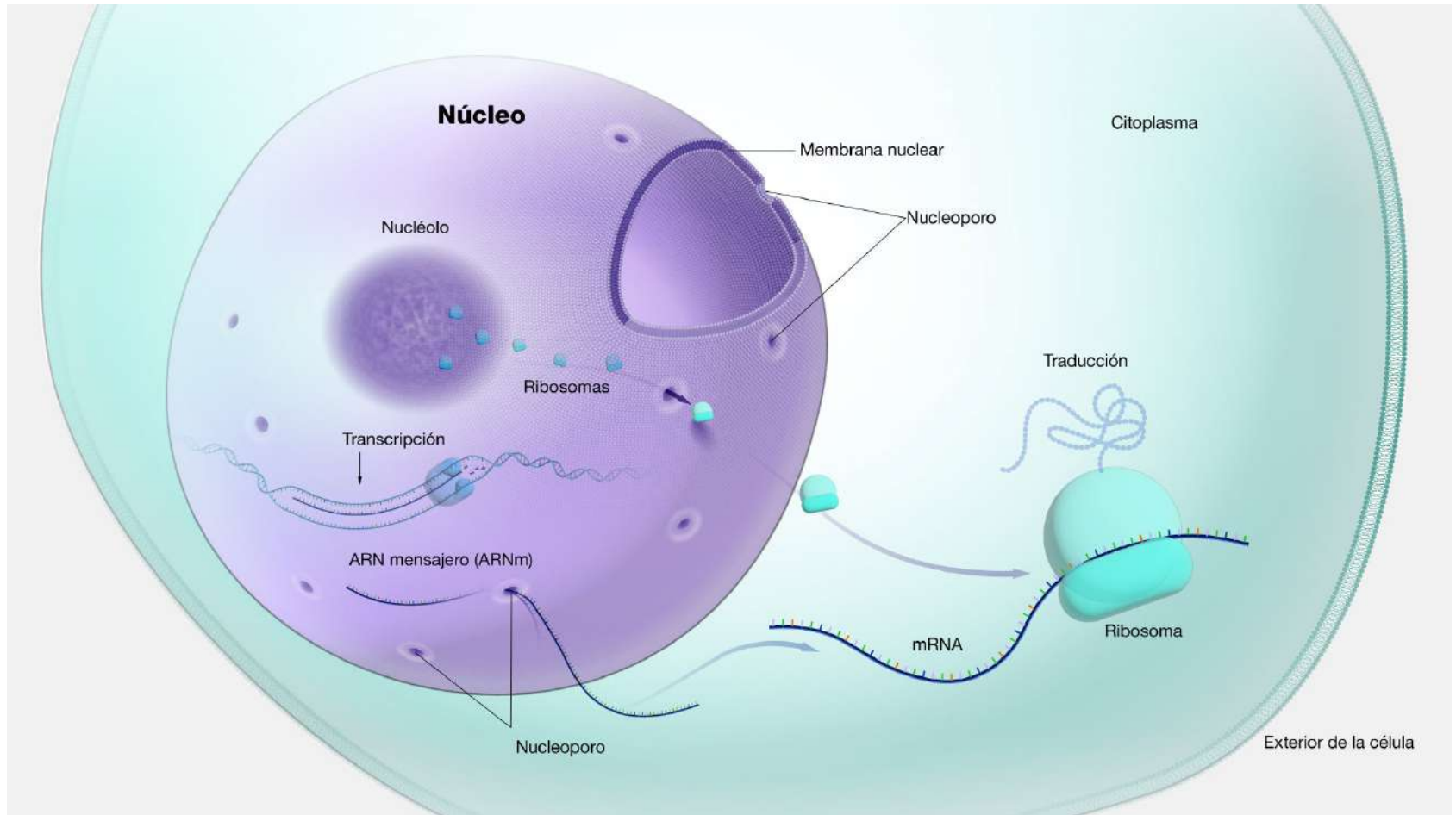


Aquí se muestran los cromosomas humanos



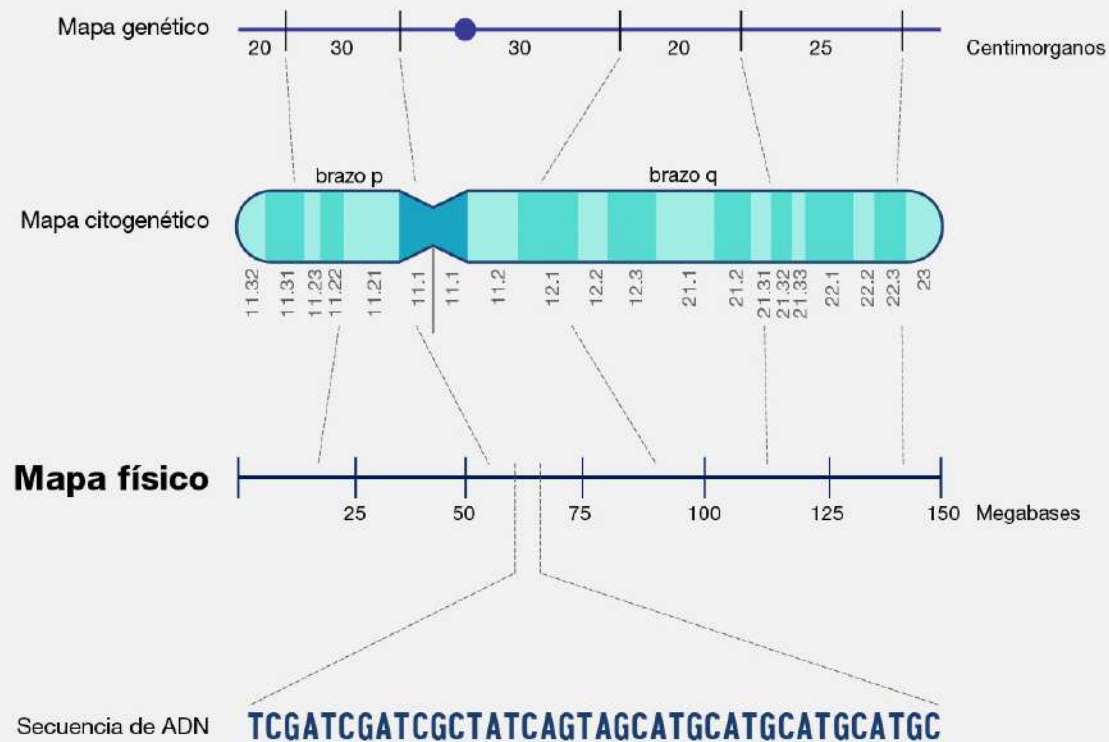
ADN mitocondrial

Núcleo celular



<https://youtu.be/JHv5lrqly-k>

Mapa genómico



Primer borrador del genoma humano completo



THE WHITE HOUSE **The East Room** For Immediate Release

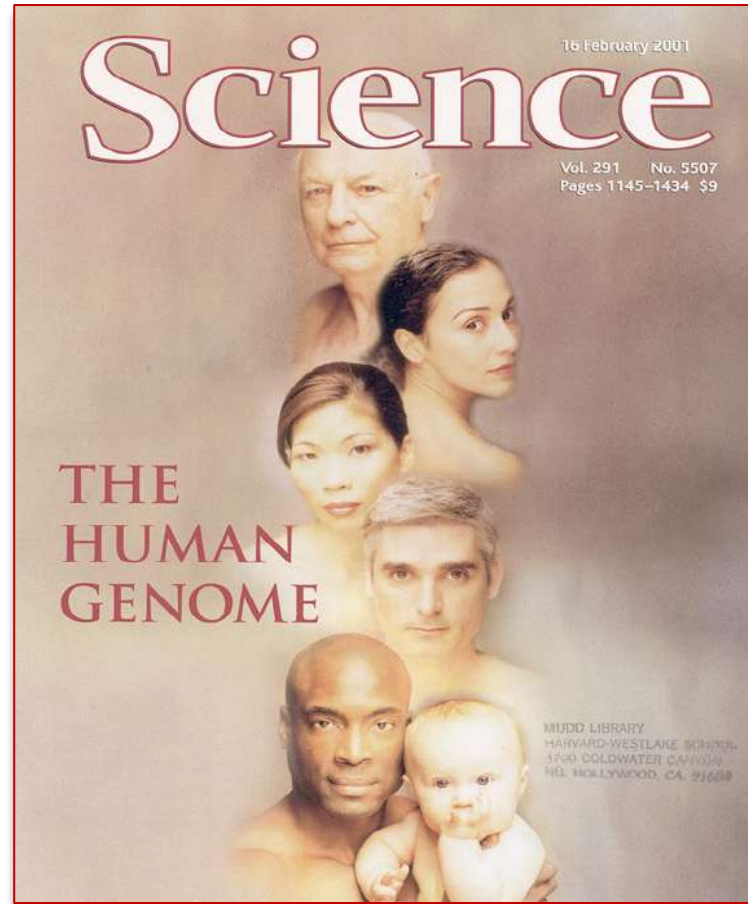
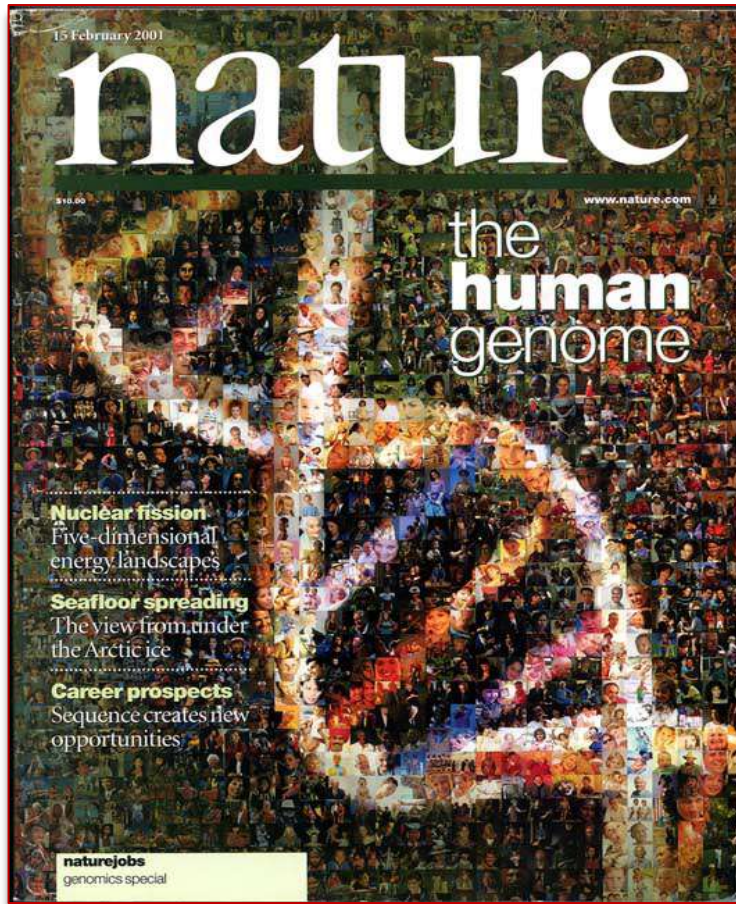
June 26, 2000 10:19 A.M. EDT

Office of the Press Secretary Remarks Made by the President, Prime Minister Tony Blair of England (via satellite), Dr. Francis Collins, Director of the National Human Genome Research Institute, and Dr. Craig Venter, President and Chief Scientific Officer, Celera Genomics Corporation, on the Completion of the First Survey of the Entire Human Genome Project

Human Genome Announcement at the White House (2000)

<https://www.youtube.com/watch?v=slRyGLmt3qc>

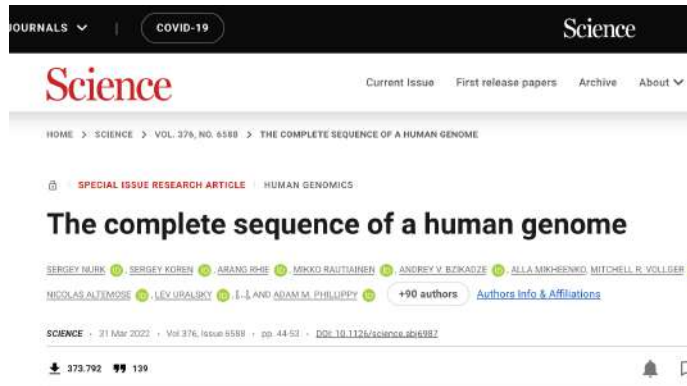
El genoma humano



Versiones del genoma humano

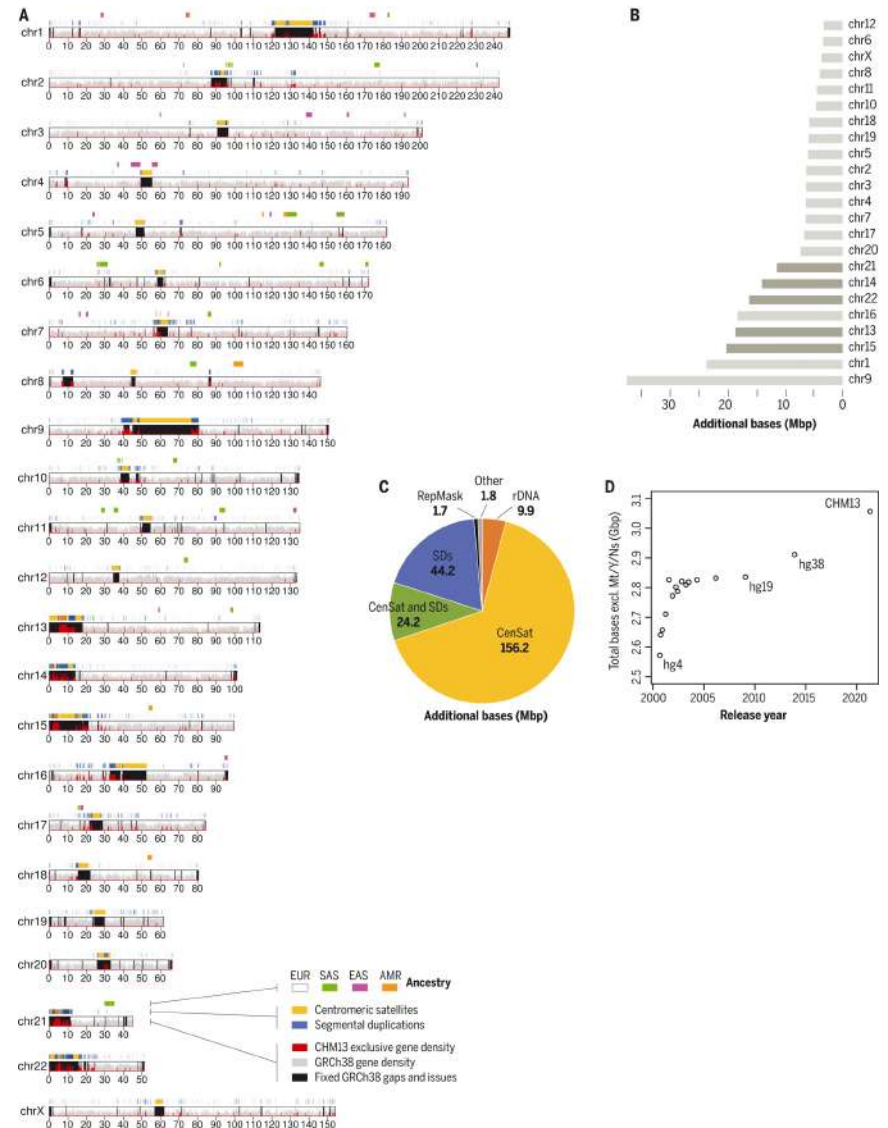
- **hg19** (también conocido como **GRCh37**)
 - Publicada en 2013
 - Esta versión todavía se utiliza ampliamente en la investigación, pero ha sido reemplazada en gran medida por la versión más reciente, hg38.
- **hg38** (también conocido como **GRCh38**)
 - Publicada en 2013.
 - Esta versión tiene una mejor calidad y una cobertura más completa que hg19, y se ha utilizado en muchos estudios genómicos recientes.
- “Telomere-to-Telomere” (**T2T**),
 - Publicada en 2022
 - Incluye secuencias de ADN que faltan en hg38, y se espera que sea una herramienta útil para la investigación genómica en el futuro.

GRCh38 vs T2T



Abstract

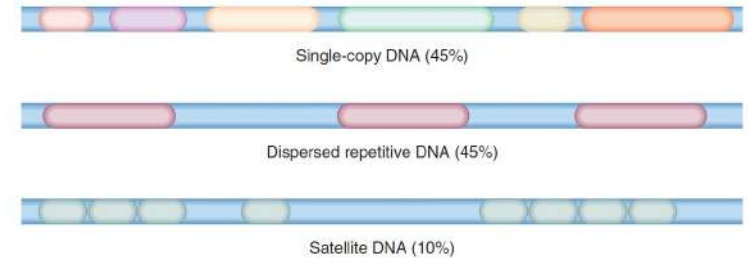
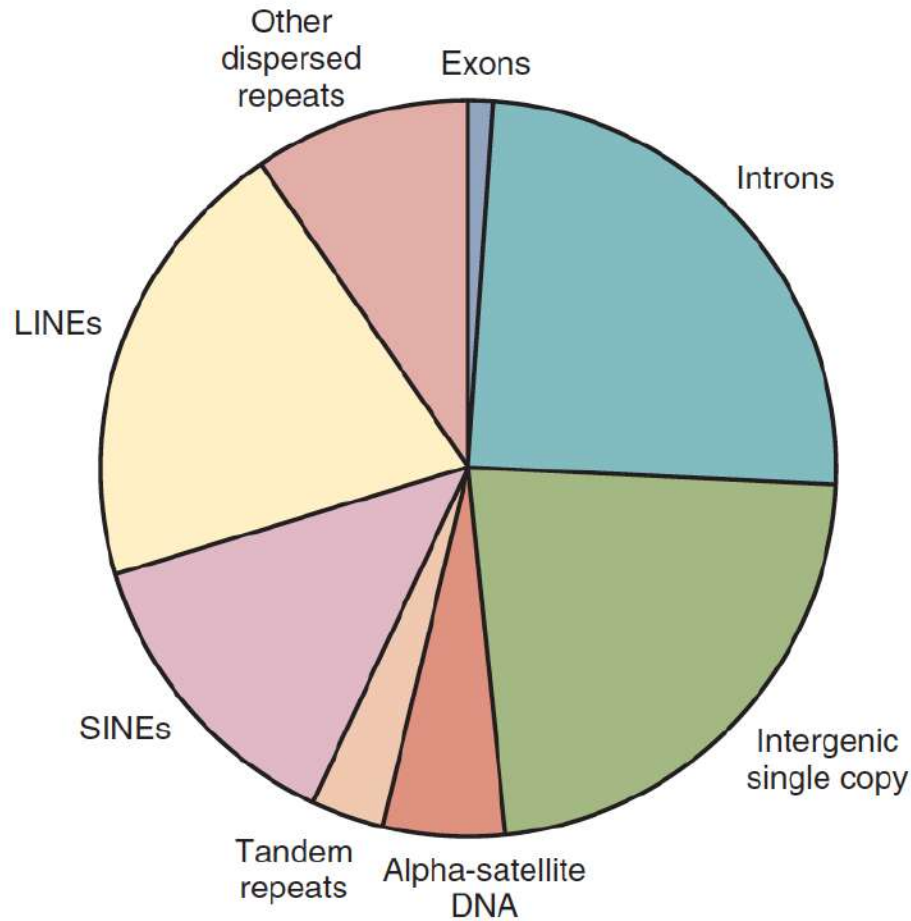
Since its initial release in 2000, the human reference genome has covered only the euchromatic fraction of the genome, leaving important heterochromatic regions unfinished. Addressing the remaining 8% of the genome, the Telomere-to-Telomere (T2T) Consortium presents a complete 3.055 billion–base pair sequence of a human genome, T2T-CHM13, that includes gapless assemblies for all chromosomes except Y, corrects errors in the prior references, and introduces nearly 200 million base pairs of sequence containing 1956 gene predictions, 99 of which are predicted to be protein coding. The completed regions include all centromeric satellite arrays, recent segmental duplications, and the short arms of all five acrocentric chromosomes, unlocking these complex regions of the genome to variational and functional studies.



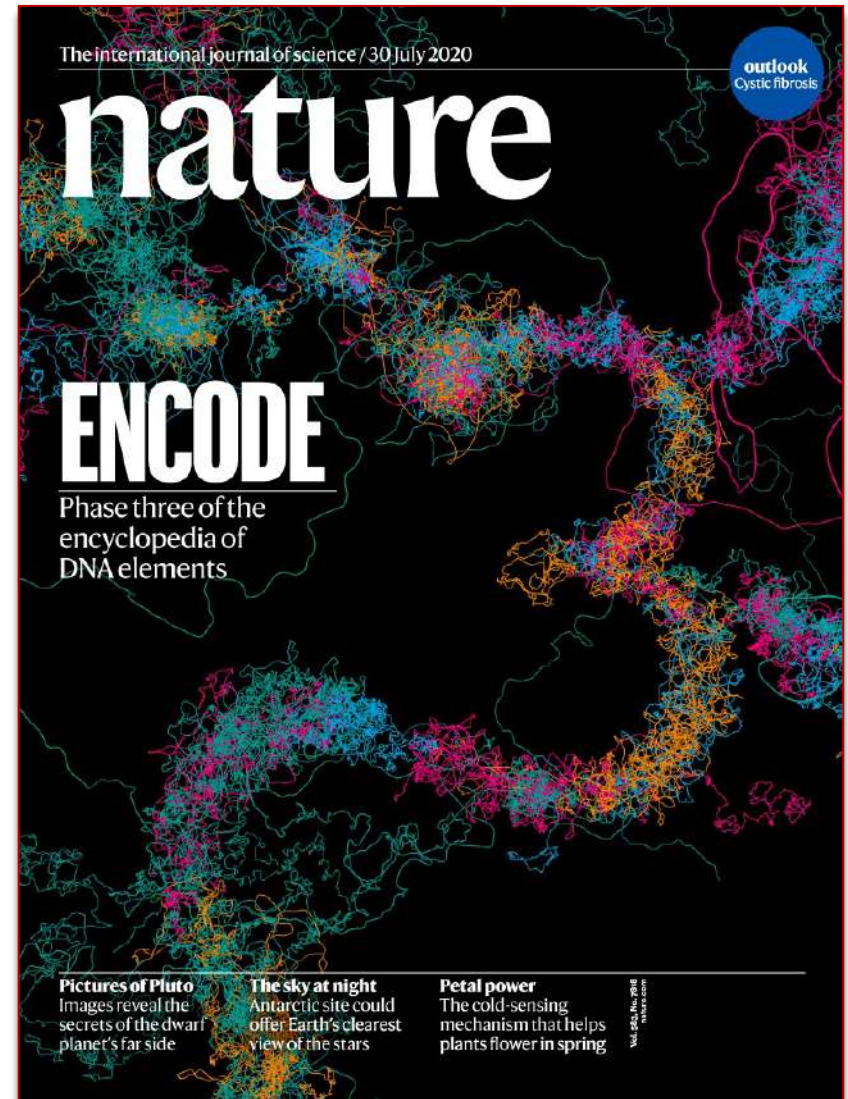
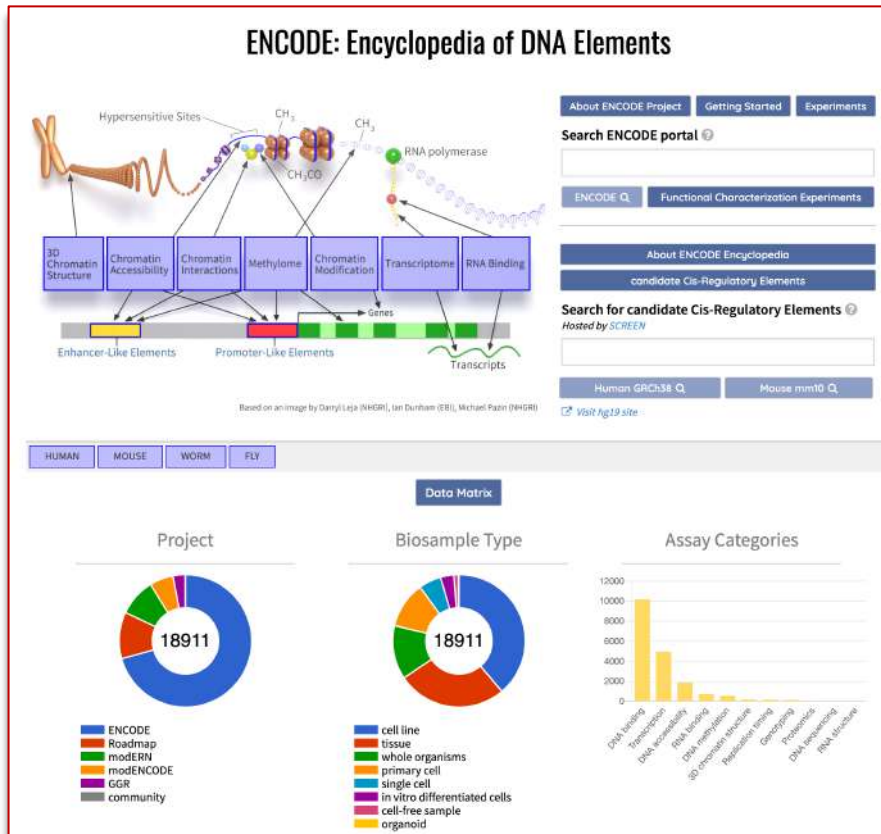
Genoma completos



Composición del genoma humano



El proyecto ENCODE



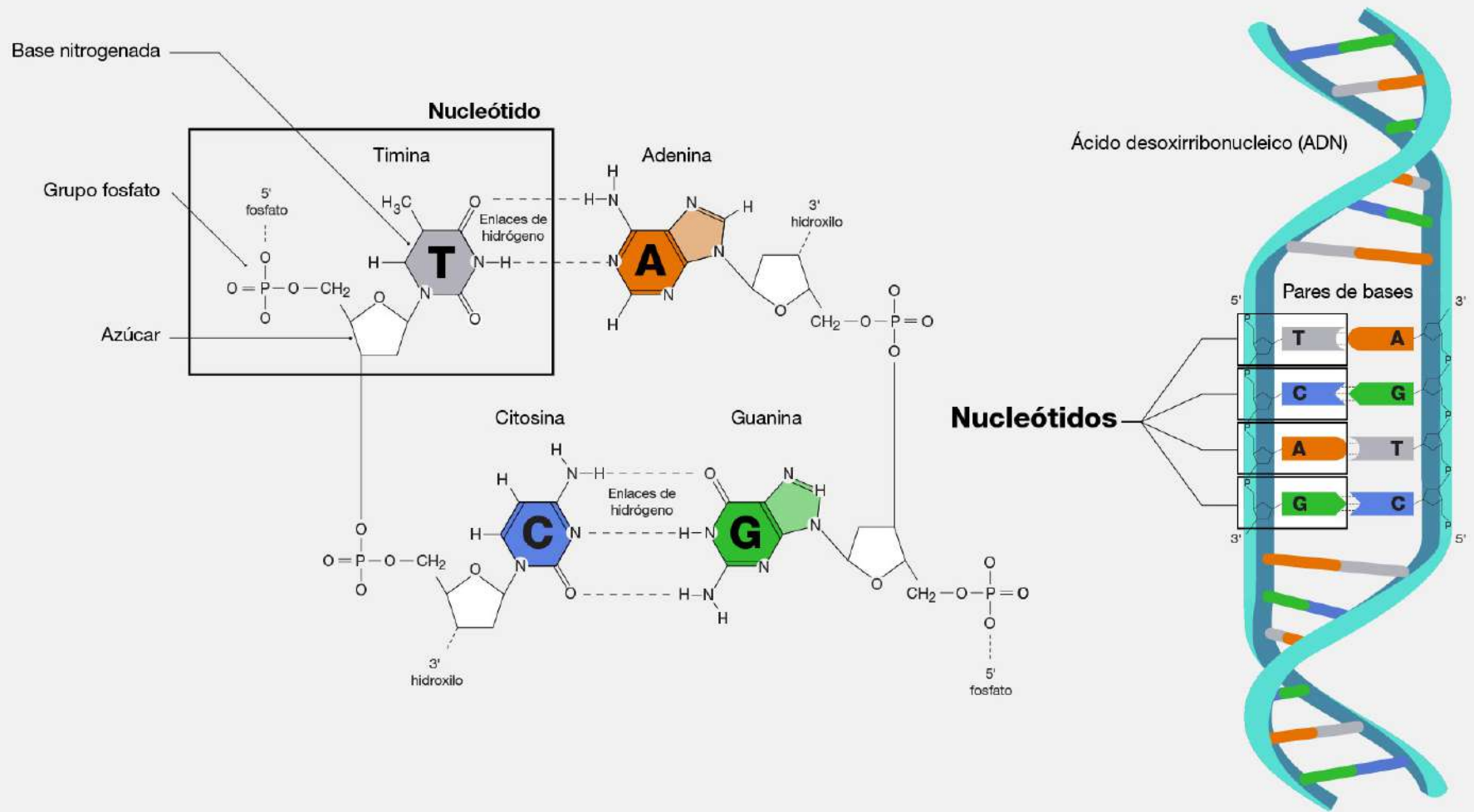
Práctica: “ver” el genoma humano

- Vamos a ver en el NCBI como podemos acceder a la secuencia del genoma humano
- También descargaremos el genoma completo desde las páginas del UCSC Genome Browser
- Podremos ver el “aspecto” del genoma humano de forma similar a como Riccardo Sabatini hacía en el vídeo de TED

Índice

- Células, genomas y cromosomas
- DNA
- Replicación
- Genes, transcripción y proteínas
- Mutaciones
- Genotipo y fenotipo

La cadena de DNA



La doble cadena de DNA en Python

```
def comp(dna) :
    compl = {'A':'T', 'T':'A', 'C':'G', 'G':'C'}
    dna = dna.upper()
    dna_c = ""
    for c in dna :
        dna_c += compl[c]
    return dna_c

dna1 = 'CATGAAGTGTGTTGGTGGTTATTG'
dna2 = comp(dna1)

print ('El DNA es: %s \n' % dna1)
print ('La cadena doble sería: ')
print ('---->')
print (dna1)
print ('|'*len(dna1))
print (dna2)
print (' '*len(dna1)-5, '<----')

print ('\nPero normalmente escribiríamos')
print ('la inversa complementaria en')
print ('el sentido de lectura correcto: ')
print (dna2[::-1])
```

El DNA es: CATGAAGTGTGTTGGTGGTTATTG

La cadena doble sería:

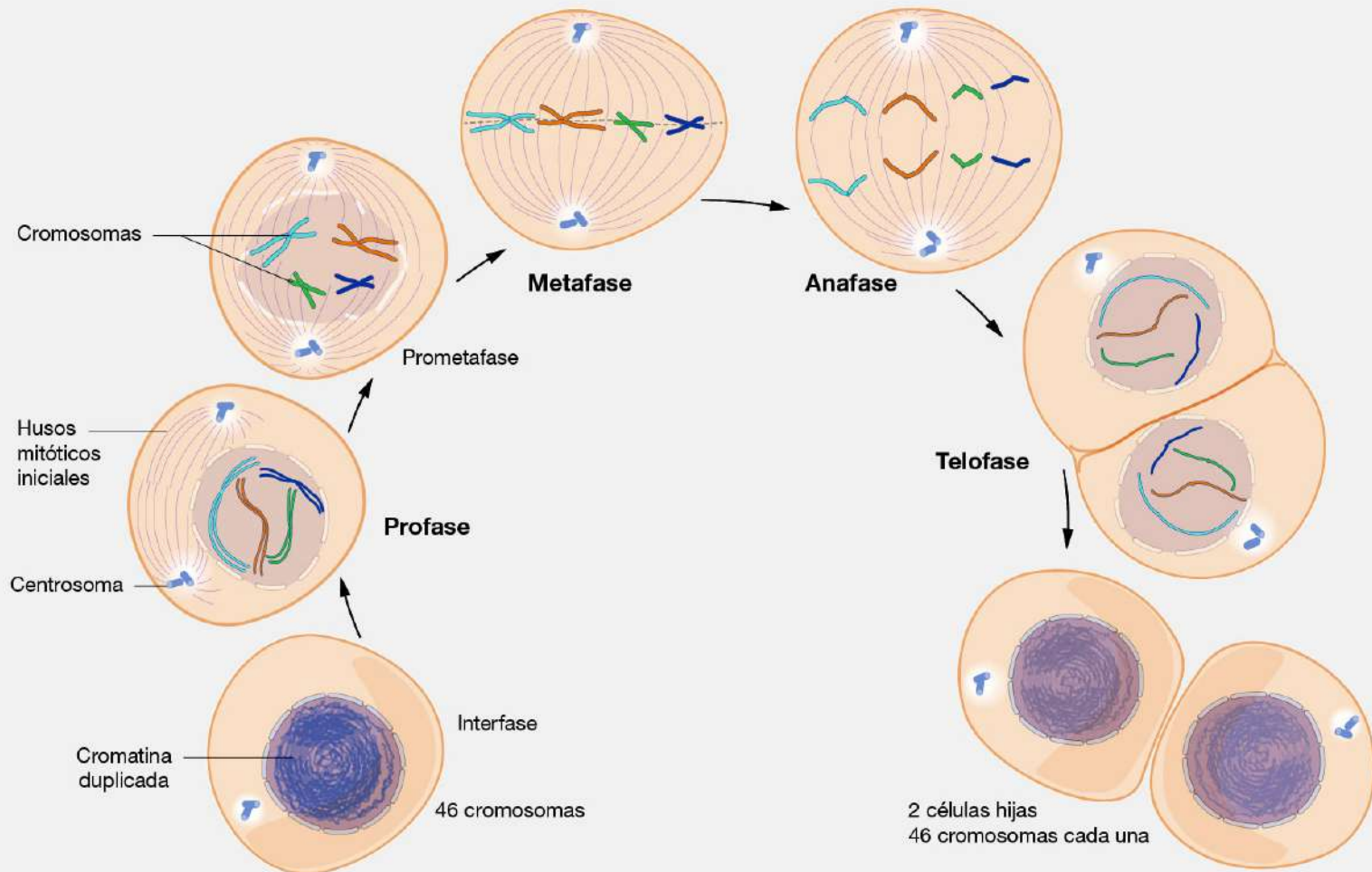
```
---->
CATGAAGTGTGTTGGTGGTTATTG
|||||
GTACTTCACACAACCACCAATAAC
<----
```

Pero normalmente escribiríamos
la inversa complementaria en
el sentido de lectura correcto:
CAATAACCACCAACACACTTCATG

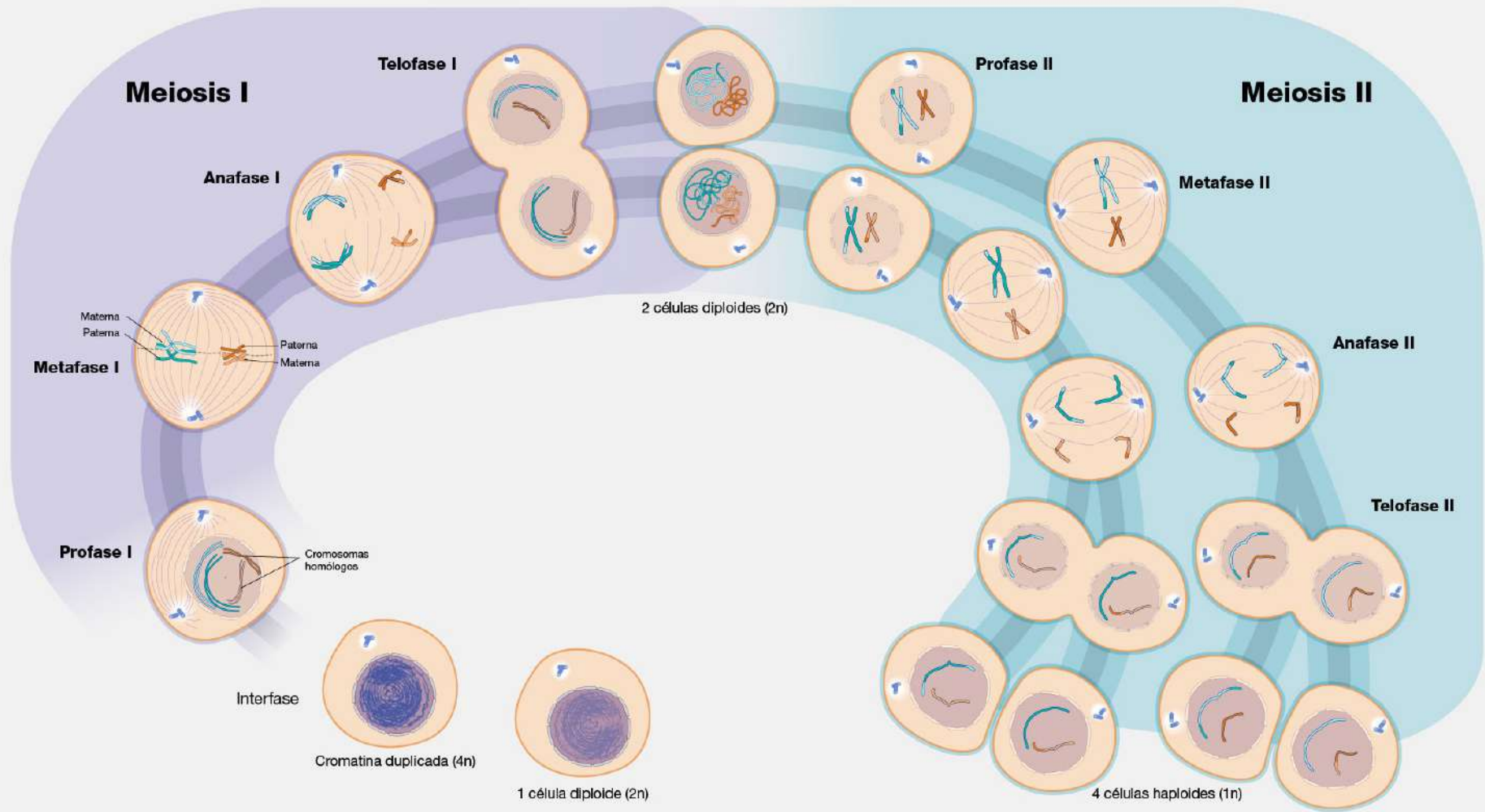
Índice

- Células, genomas y cromosomas
- DNA
- Replicación
- Genes, transcripción y proteínas
- Mutaciones
- Genotipo y fenotipo

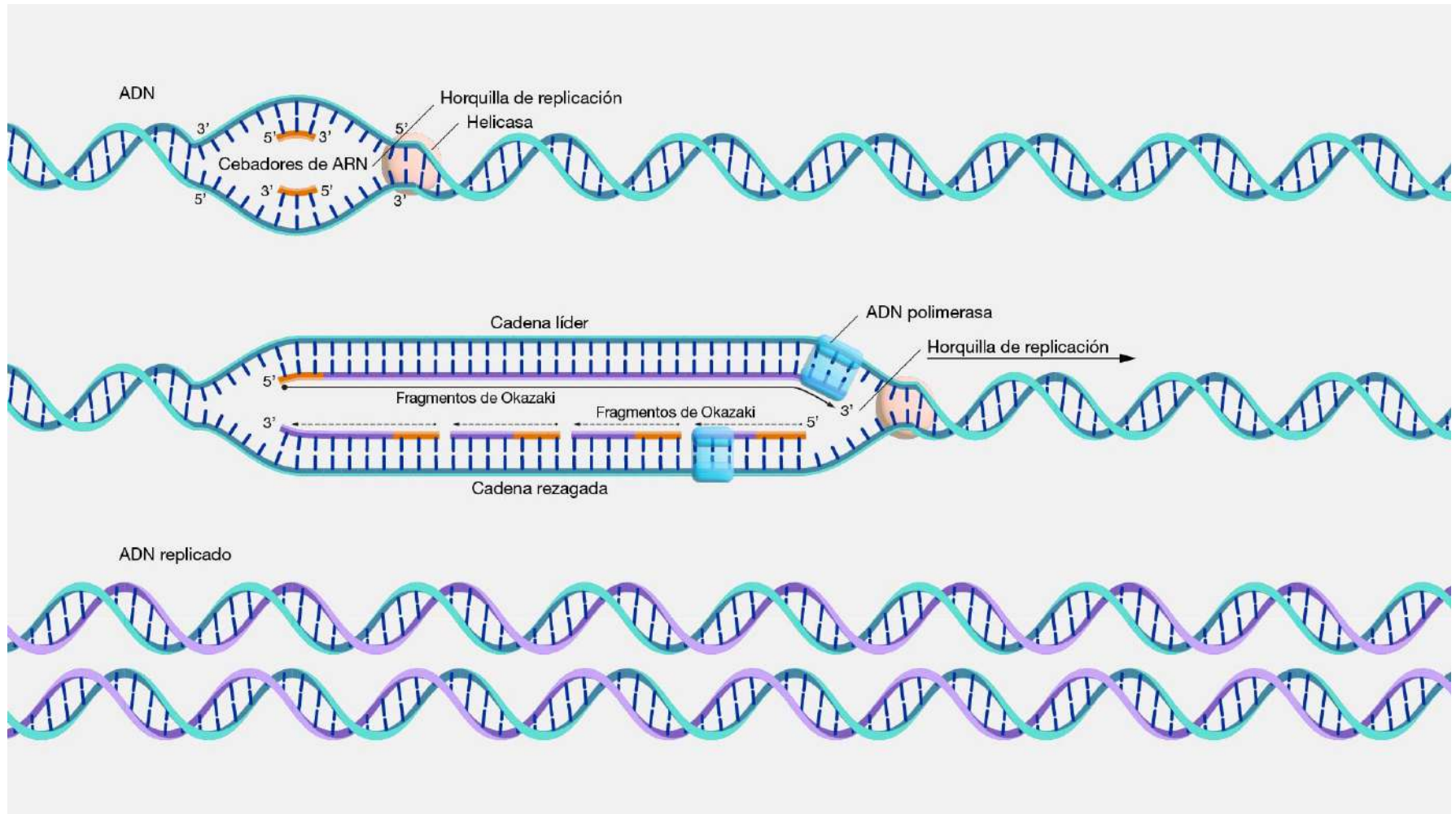
Mitosis



Meiosis



Replicación del ADN



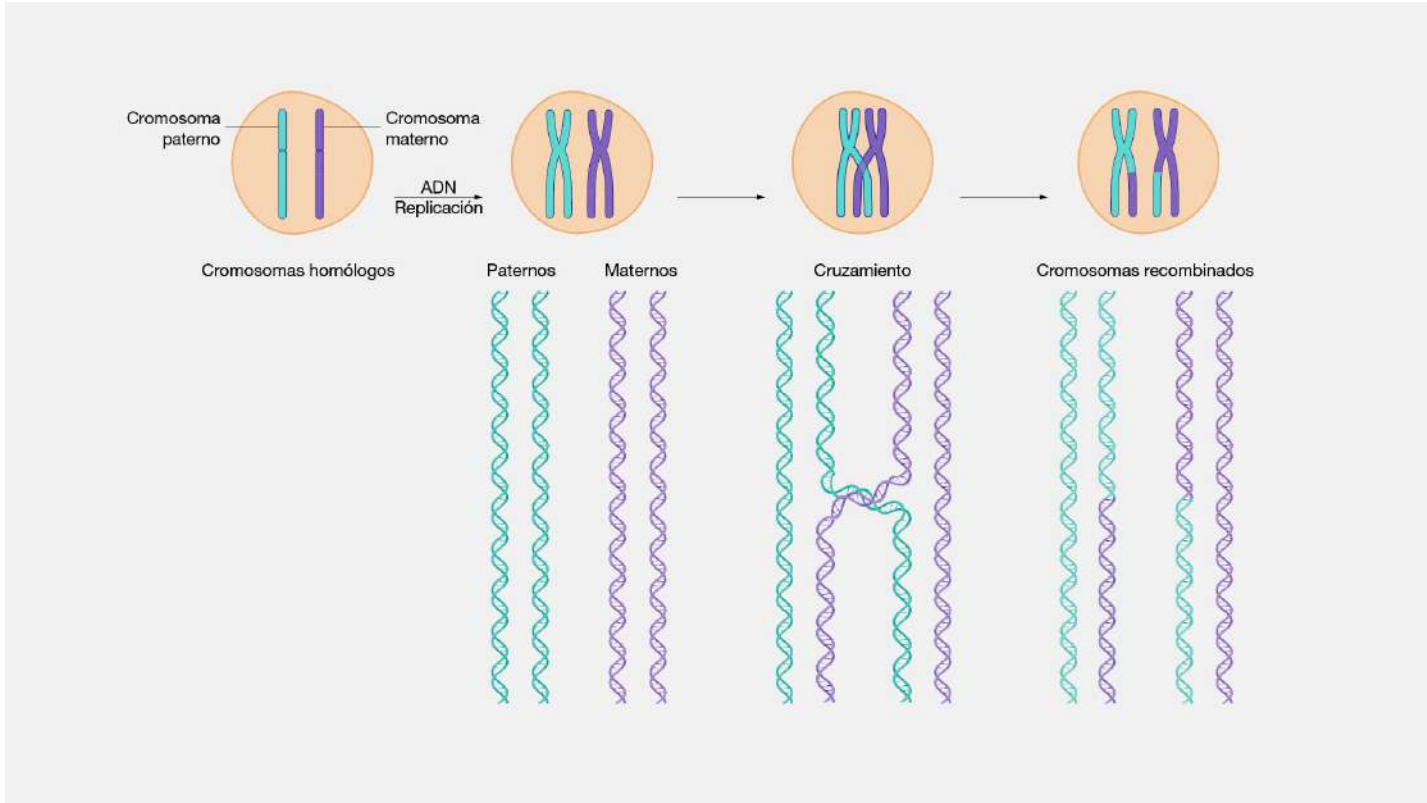
Replicación del ADN

1. División celular
2. Desarrollo embrionario
3. Regeneración de tejidos
4. Reproducción sexual
5. Reparación del ADN

Recombinación homóloga

Durante la formación de los óvulos y los espermatozoides (meiosis), pares de cromosomas del progenitor masculino y el femenino se alinean de forma que secuencias parecidas de ADN puedan cruzarse, o intercambiarse, de un cromosoma al otro.

Ese intercambio de ADN es una fuente importante de la variación genómica que vemos en la descendencia.



Células: Línea germinal

Línea germinal (células germinales)

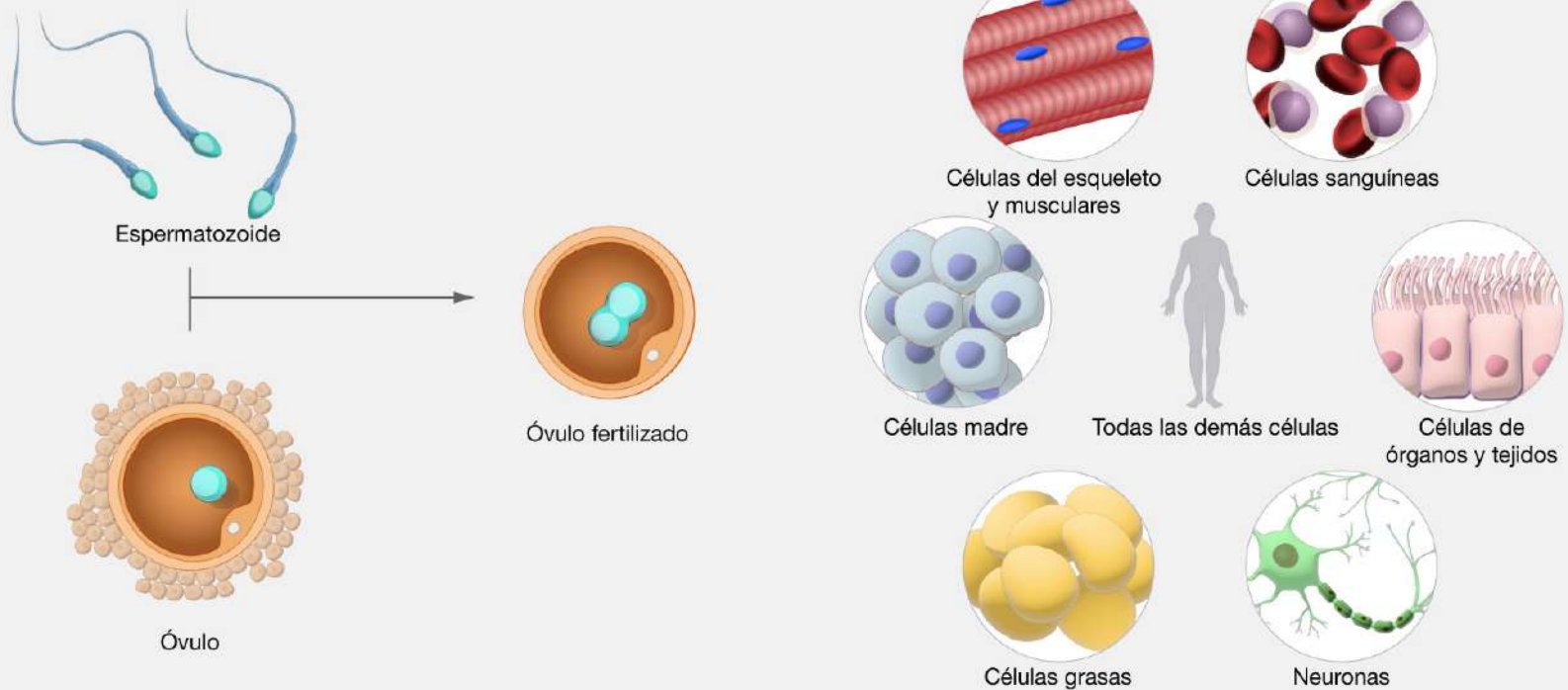
Haploide

23 cromosomas (n) en seres humanos

Células somáticas

Diploide

46 cromosomas (2n) en seres humanos



Células: somáticas

Línea germinal (células germinales)

Haploide

23 cromosomas (n) en seres humanos

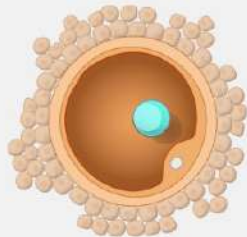
Células somáticas

Diploide

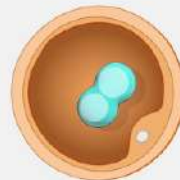
46 cromosomas (2n) en seres humanos



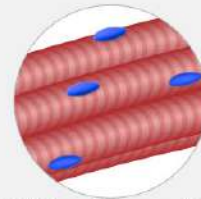
Espermatozoide



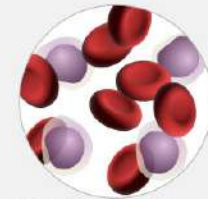
Óvulo



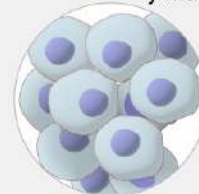
Óvulo fertilizado



Células del esqueleto y musculares



Células sanguíneas



Células madre



Todas las demás células



Células de órganos y tejidos



Células grasas



Neuronas

Índice

- Células, genomas y cromosomas
- DNA
- Replicación
- Genes, transcripción y proteínas
- Mutaciones
- Genotipo y fenotipo

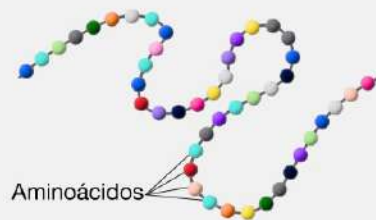
Proteínas

- Son macromoléculas formadas por cadenas lineales de aminoácidos
- La secuencia de aminoácidos que dan lugar a una proteína viene determinada por la secuencia de nucleótidos en el gen correspondiente que lo codifica.
- La síntesis de proteínas tiene lugar a través de la traducción:
 - Llevada a cabo por los ribosomas
 - Guiada por la información de una molécula de mRNA que actúa como molde
- Distintas células, tejido u organismos pueden necesitar unas u otras proteínas para llevar a cabo distintas funciones y sobrevivir

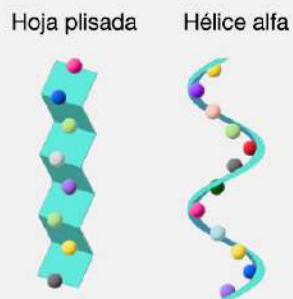
Niveles de organización de las proteínas

Niveles de organización de las proteínas

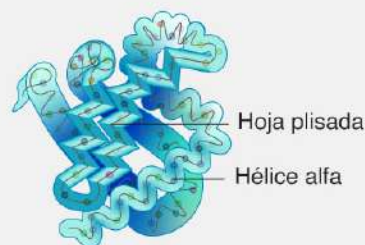
La estructura proteica primaria es la secuencia de una cadena de aminoácidos.



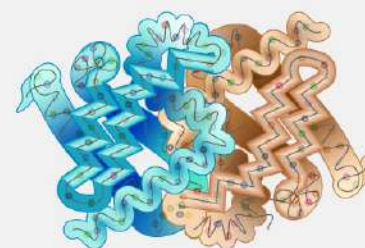
La estructura proteica secundaria se produce cuando la secuencia de aminoácidos se pliega y adopta una forma tridimensional.



La estructura proteica terciaria se produce cuando una proteína madura se pliega sobre sí misma.



La estructura proteica cuaternaria es una proteína que consta de más de una cadena polipeptídica.



Protein Data Bank

RCSB PDB | Deposit | Search | Visualize | Analyze | Download | Learn | More | Documentation | Careers | MyPDB | Contact us

RCSB PDB PROTEIN DATA BANK

201,515 Structures from the PDB
1,068,577 Computed Structure Models (CSM)

3D Structures | Enter search term(s), Entry ID(s), or sequence | Include CSM | Help

Advanced Search | Browse Annotations

PDB-101 | wwPDB | EMDDataResource | NUCLEIC ACID DATABASE | wwPDB Foundation

f t y

New: More Computed Structure Models (CSM) available | Learn more

Welcome

Deposit

Search

Visualize

Analyze

Download

Learn

RCSB Protein Data Bank (RCSB PDB) enables breakthroughs in science and education by providing access and tools for exploration, visualization, and analysis of:

- Experimentally-determined 3D structures from the **Protein Data Bank (PDB)** archive
- Computed Structure Models (CSM)** from AlphaFold DB and ModelArchive

These data can be explored in context of external annotations providing a structural view of biology.

COVID-19 CORONAVIRUS Resources

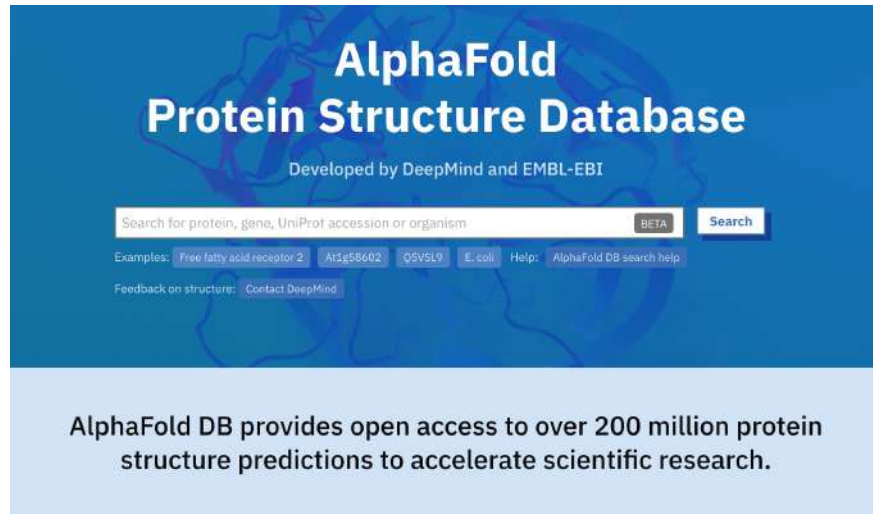
200000 Structures in the Protein Data Bank

February Molecule of the Month

SARS-CoV-2 Nucleocapsid and Home Tests

<https://www.rcsb.org/>

AlphaFold



The image shows the homepage of the AlphaFold Protein Structure Database. The header features the title "AlphaFold Protein Structure Database" in white text on a blue background, with "Developed by DeepMind and EMBL-EBI" below it. A search bar with a "BETA" label and a "Search" button is present. Below the search bar, there are examples of protein IDs (Free fatty acid receptor 2, A1158602, Q5VSL9, E. coli) and a link to the AlphaFold DB search help. A feedback link "Contact DeepMind" is also visible. The main body of the page has a light blue background with the text: "AlphaFold DB provides open access to over 200 million protein structure predictions to accelerate scientific research."

AlphaFold Protein Structure Database

Developed by DeepMind and EMBL-EBI

Search for protein, gene, UniProt accession or organism BETA Search

Examples: [Free fatty acid receptor 2](#) [A1158602](#) [Q5VSL9](#) [E. coli](#) Help: [AlphaFold DB search help](#)

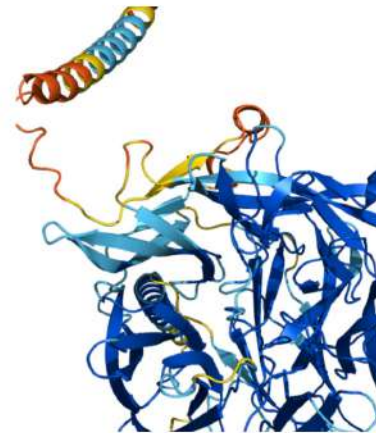
Feedback on structure: [Contact DeepMind](#)

AlphaFold DB provides open access to over 200 million protein structure predictions to accelerate scientific research.

Background

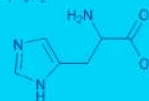
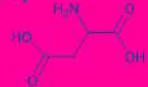
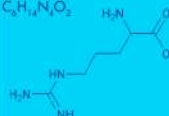
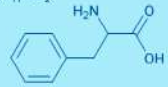
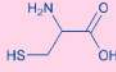
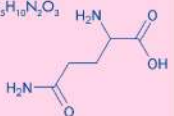
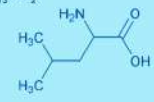
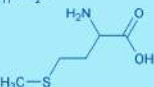
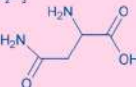
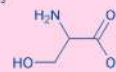
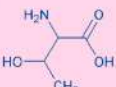
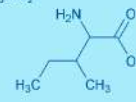
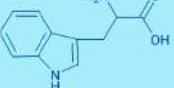
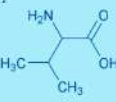
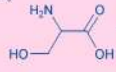
AlphaFold is an AI system developed by **DeepMind** that predicts a protein's 3D structure from its amino acid sequence. It regularly achieves accuracy competitive with experiment.

DeepMind and EMBL's European Bioinformatics Institute (**EMBL-EBI**) have partnered to create AlphaFold DB to make these predictions freely available to the scientific community. The latest database release contains over 200 million entries, providing broad coverage of **UniProt** (the standard repository of protein sequences and annotations). We provide individual **downloads** for the human proteome and for the proteomes of 47 other key organisms important in research and global health. We also provide a download for the manually curated subset of UniProt (**Swiss-Prot**).

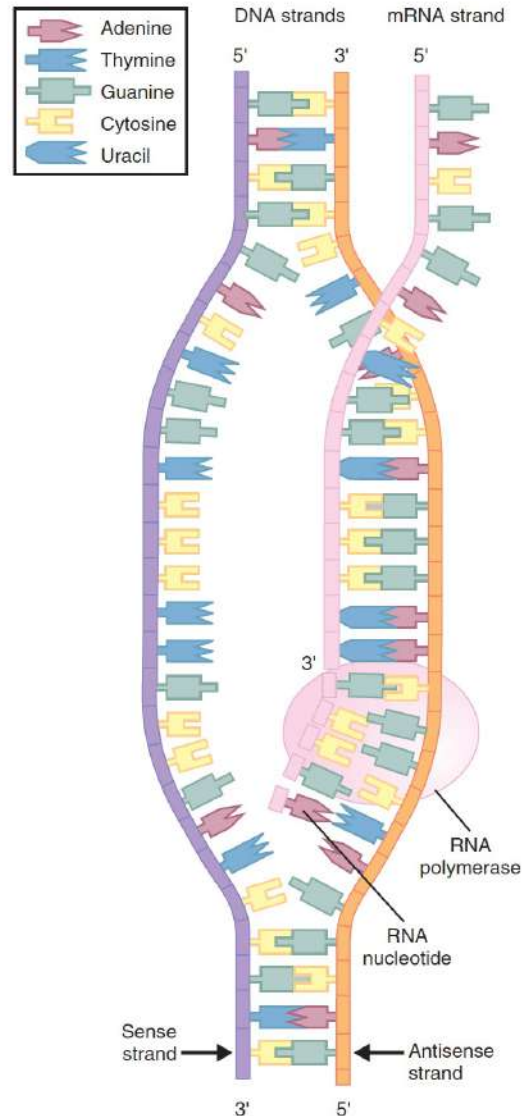


Q813H7: May protect the malaria parasite against attack by the immune system. Mean pLDDT 85.57.

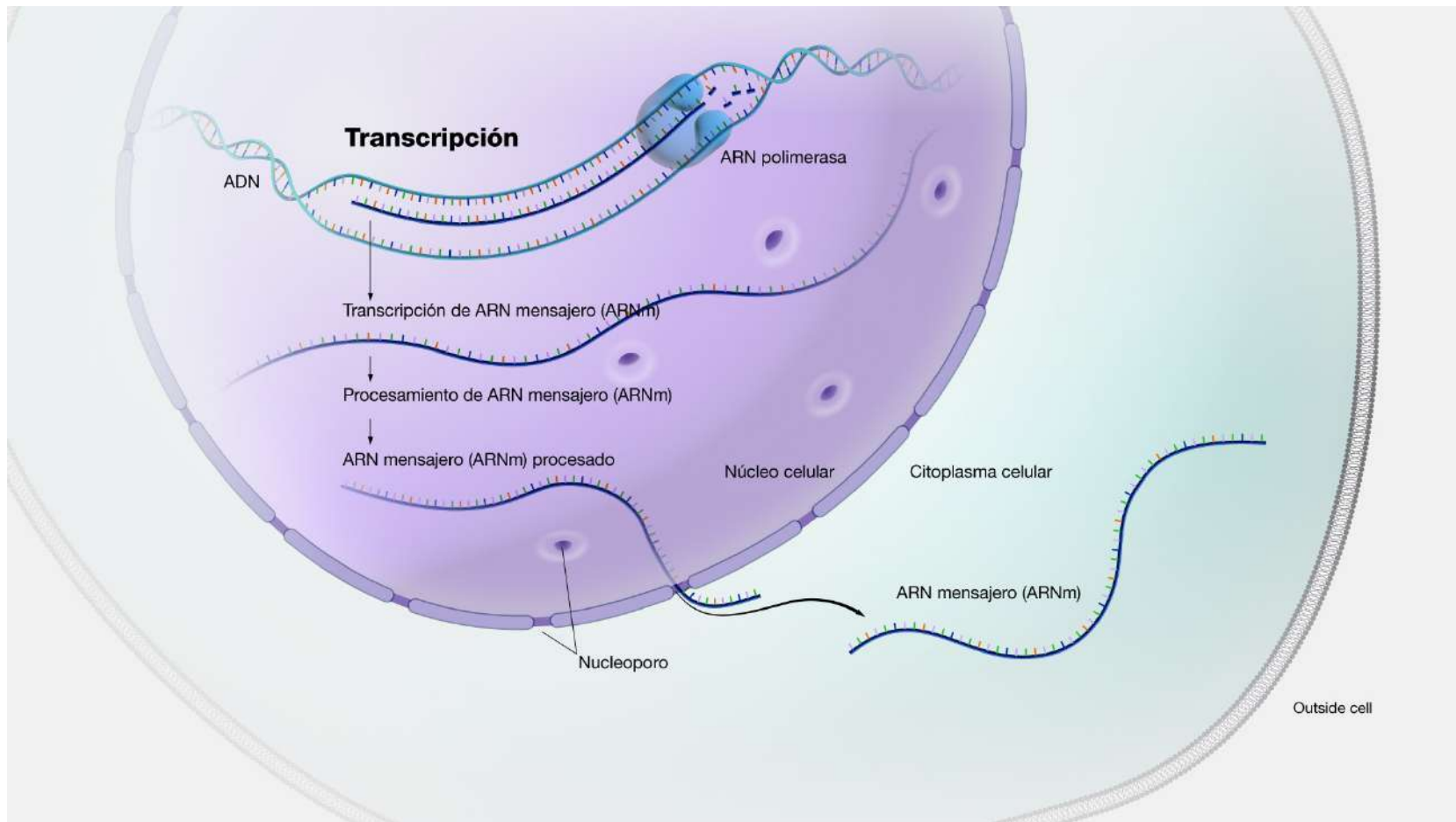
Aminoácidos

Periodic Chart of Amino Acids						www.bachem.com											
H 155.16 137.14 C₆H₉N₃O₂  His Histidine		D 133.10 115.09 C₄H₇NO₄  Asp Aspartic Acid		R 174.20 156.19 C₉H₁₄N₄O₂  Arg Arginine		F 165.19 147.18 C₉H₉NO₂  Phe Phenylalanine		A 89.09 71.08 C₃H₇NO₂  Ala Alanine		C 121.16 103.14 C₃H₇NO₂S  Cys Cysteine		G 75.07 57.05 C₂H₃NO₂  Gly Glycine		Q 146.15 128.13 C₅H₁₀N₂O₃  Gln Glutamine		E 147.13 129.11 C₇H₉NO₄  Glu Glutamic Acid	
K 146.19 128.17 C₆H₁₄N₃O₂  Lys Lysine		L 131.18 113.16 C₆H₁₃NO₂  Leu Leucine		M 149.21 131.20 C₅H₁₁NO₂S  Met Methionine		N 132.12 114.10 C₄H₈N₂O₃  Asn Asparagine		S 105.09 87.08 C₃H₇NO₃  Ser Serine		Y 181.19 163.17 C₉H₉NO₃  Tyr Tyrosine		T 119.12 101.10 C₄H₉NO₃  Thr Threonine					
I 131.18 113.16 C₆H₁₃NO₂  Ile Isoleucine		W 204.23 186.21 C₁₁H₁₂N₂O₂  Trp Tryptophan		P 115.13 97.12 C₅H₉NO₂  Pro Proline		V 117.15 99.13 C₅H₁₁NO₂  Val Valine		Basic Non-polar (hydrophobic) Polar, uncharged Acidic 1-Letter Amino Acid Code Relative Molecular Mass M_r - H₂O Molecular Formula S 105.09 87.08 C₃H₇NO₃  Serine 3-Letter Amino Acid Code Chemical Structure Chemical Name									

Transcripción: DNA $\rightarrow$ mRNA

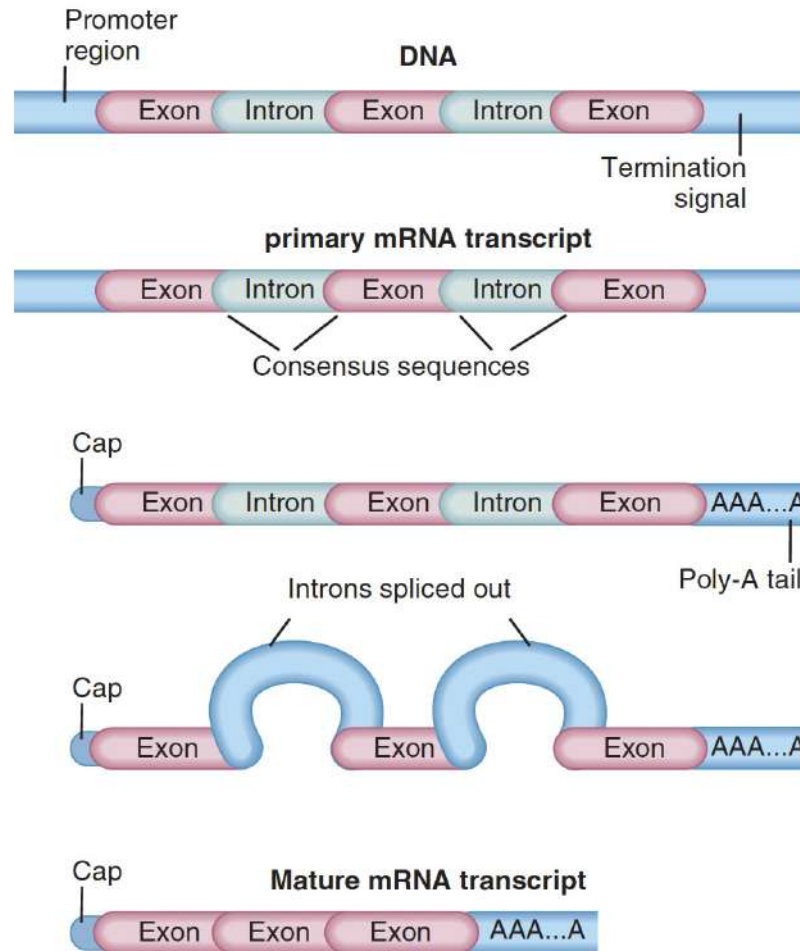


Transcripción

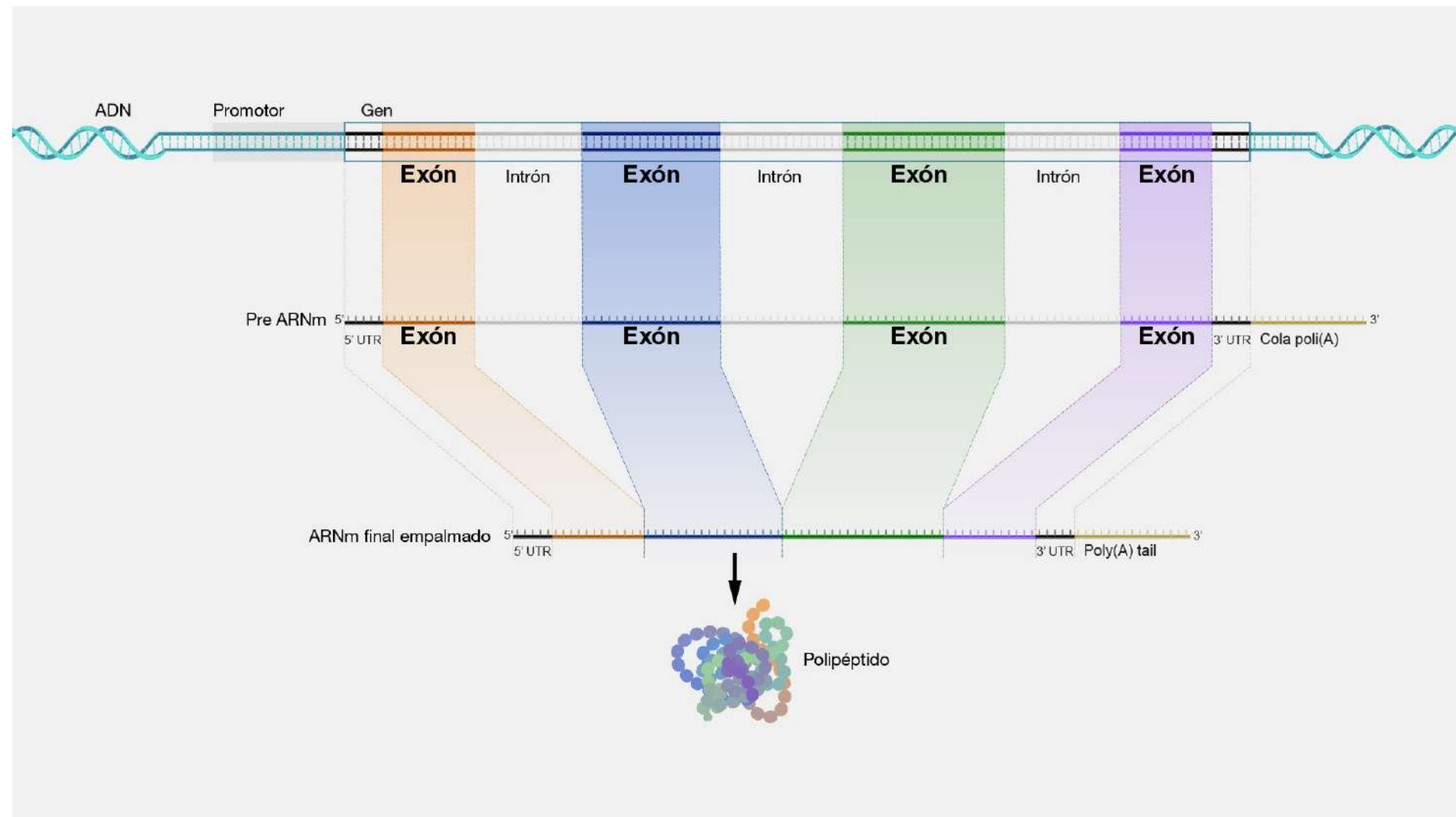


<https://youtu.be/kHlgG035Id8>

mRNA maduro y splicing

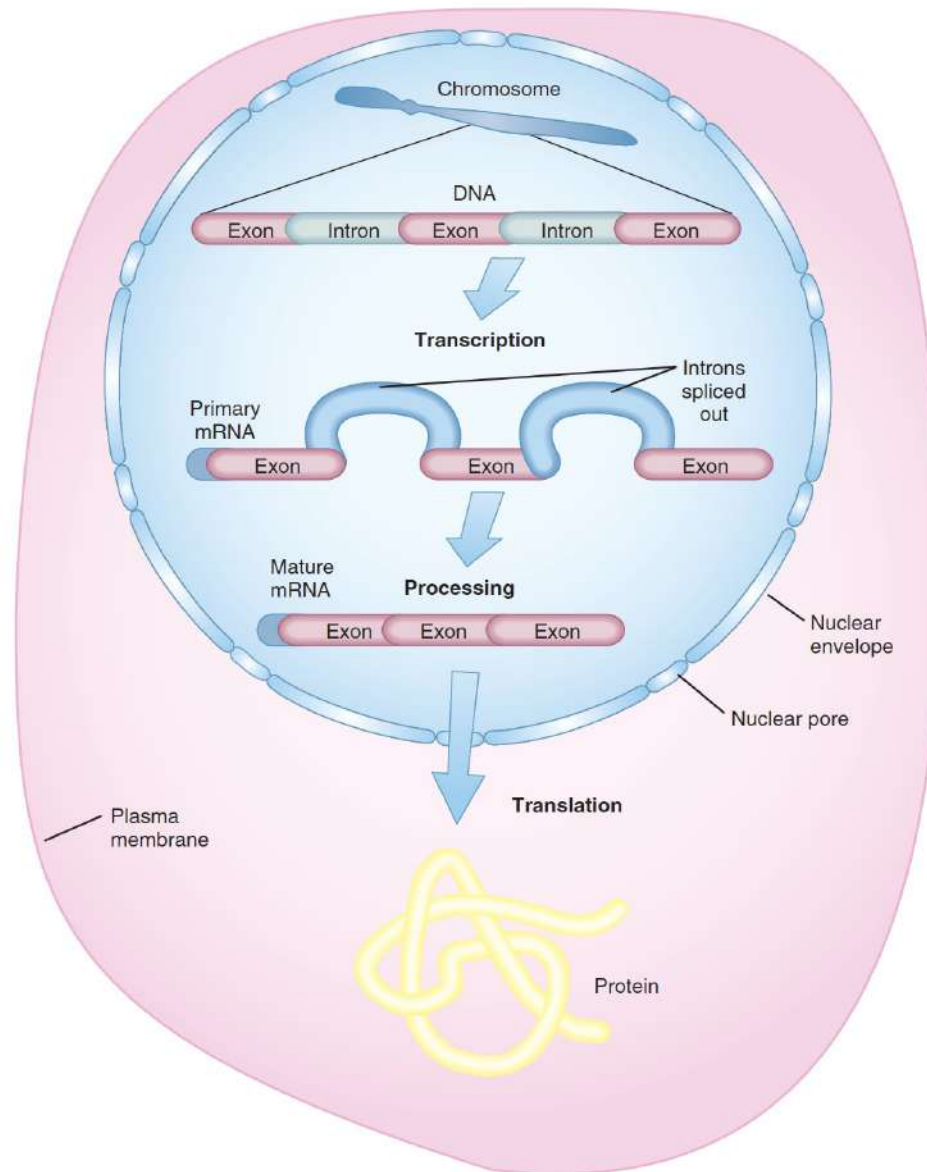


Gen, exones e intrones

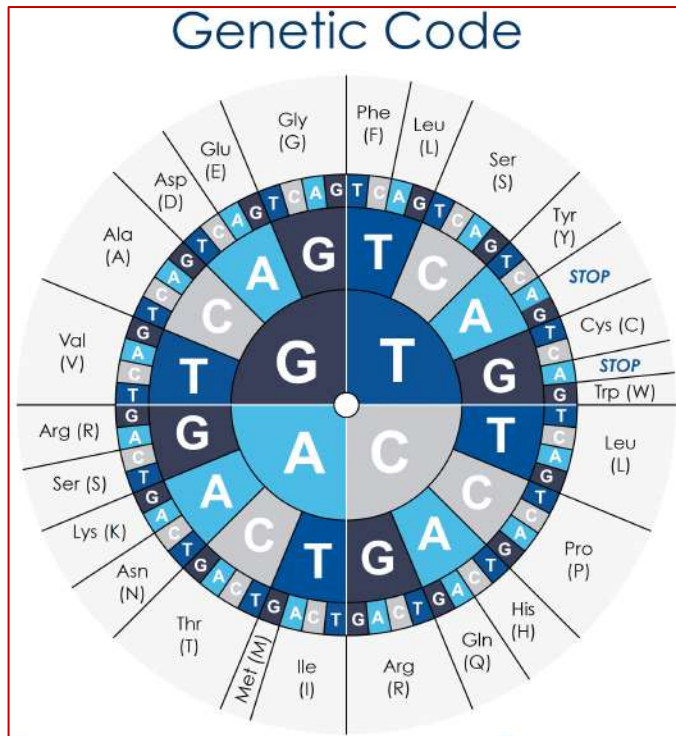


<https://youtu.be/UQtY82s7nFs>

De genes a proteínas



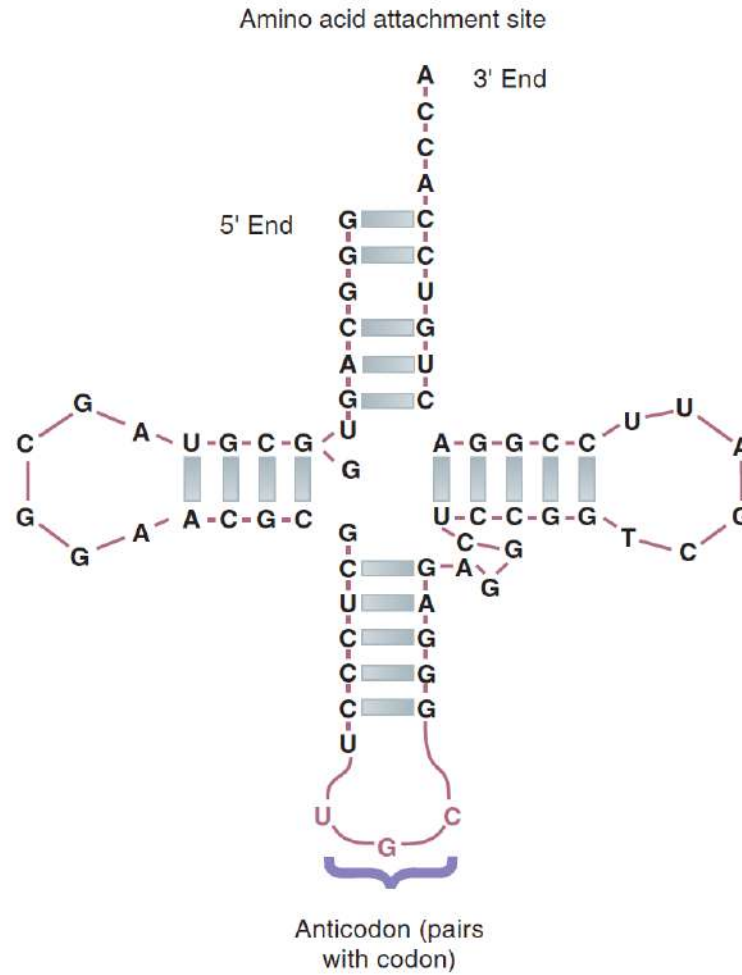
El código genético



Codon Table

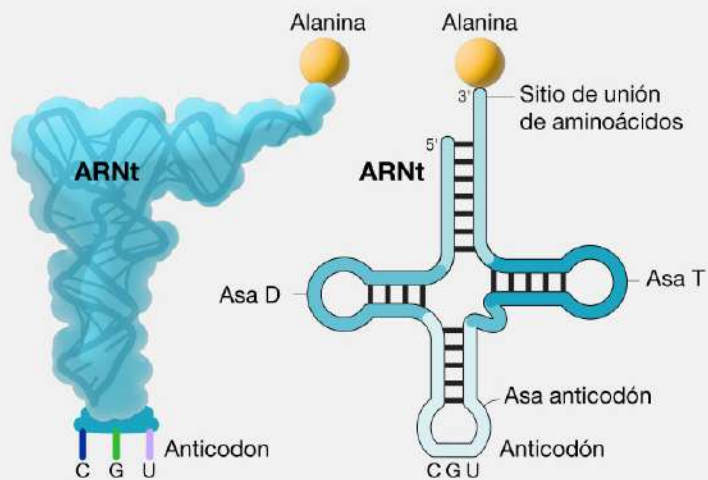
		Second Position				
		T	C	A	G	
First Position	T	TTT } Phe TTC } TTA } Leu TTG }	TCT } Ser TCC } TCA } TCG }	TAT } Tyr TAC } TAA STOP TAG STOP	TGT } Cys TGC } TGA STOP TGG Trp	T C A G
	C	CTT } Leu CTC } CTA } CTG }	CCT } Pro CCC } CCA } CCG }	CAT } His CAC } CAA } Gln CAG }	CGT } Arg CGC } CGA } CGG }	T C A G
	A	ATT } Ile ATC } ATA } Met ATG }	ACT } Thr ACC } ACA } ACG }	AAT } Asn AAC } AAA } Lys AAG }	AGT } Ser AGC } AGA } Arg AGG }	T C A G
	G	GTT } Val GTC } GTA } GTG }	GCT } Ala GCC } GCA } GCG }	GAT } Asp GAC } GAA } Glu GAG }	GGT } Gly GGC } GGA } GGG }	T C A G

tRNA: secuencia

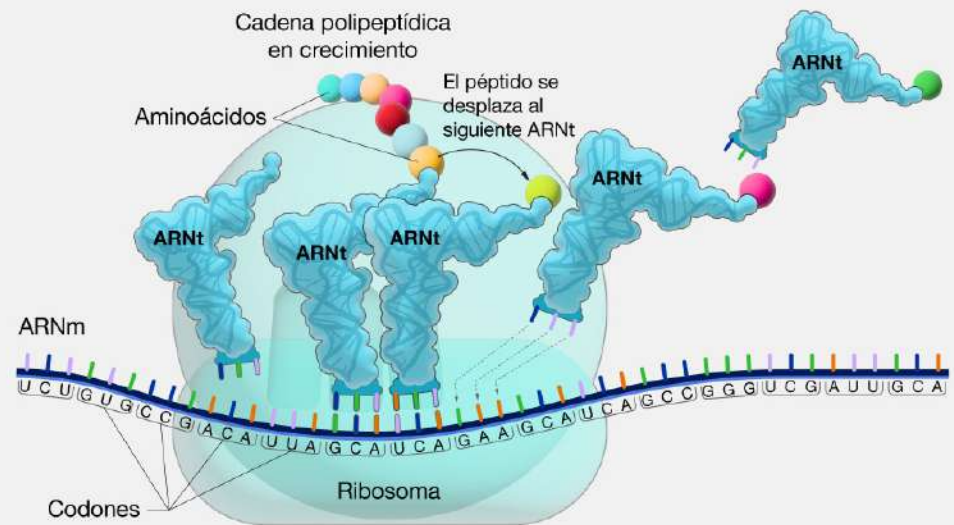


tRNA

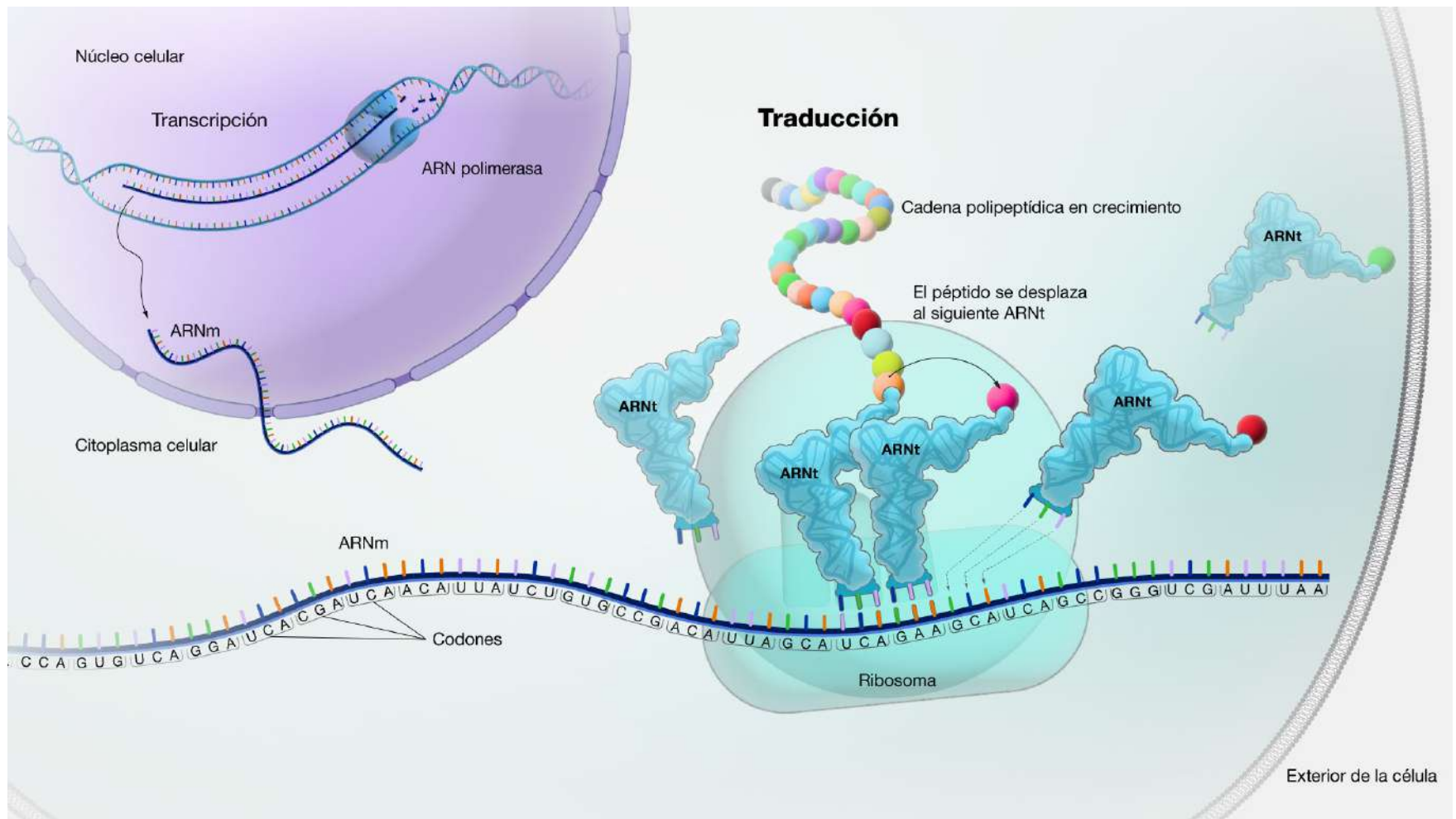
Formas habituales de representar el ARN de transferencia (ARNt)



Durante la traducción

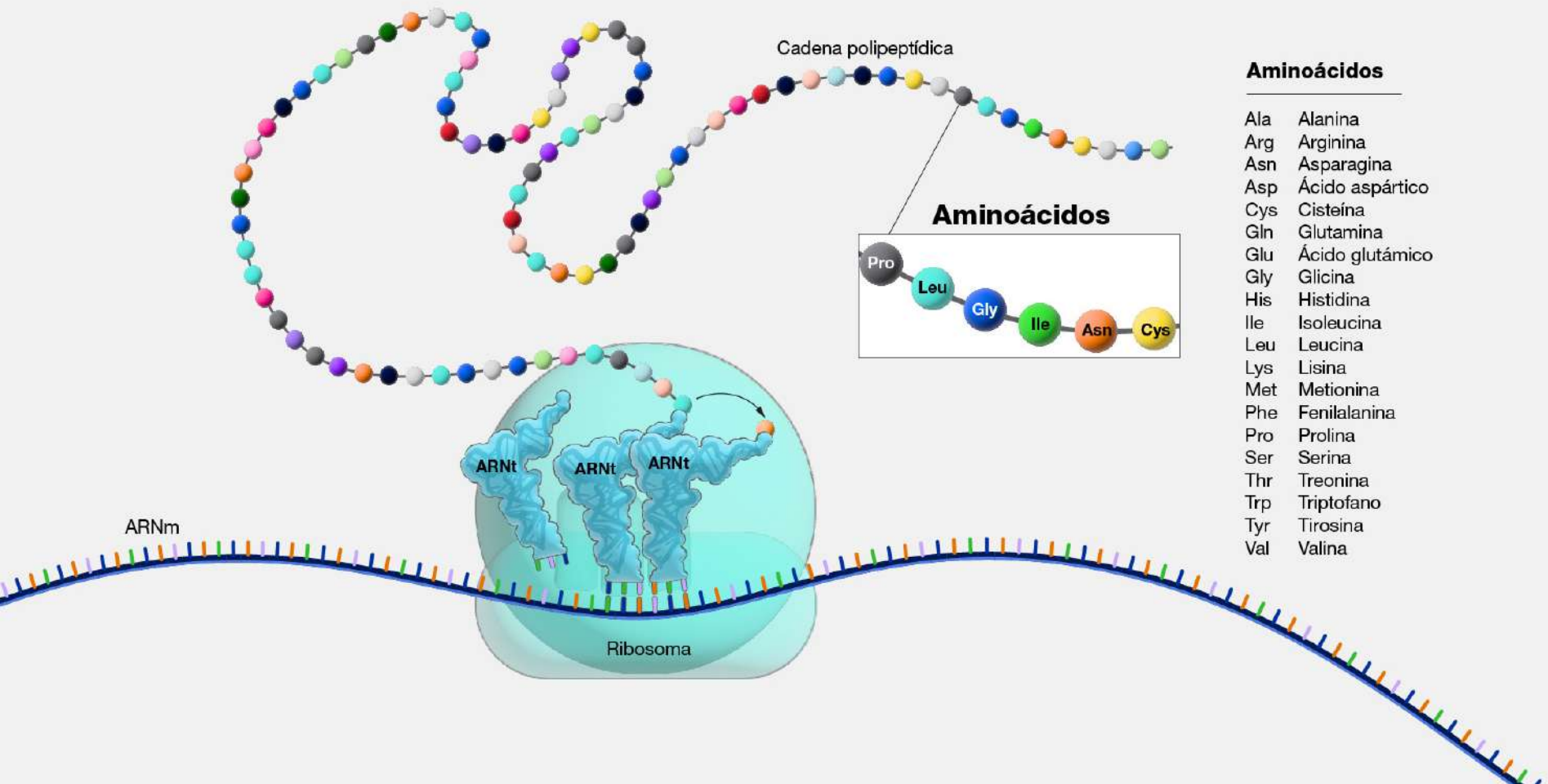


Traducción de mRNA a aminoácidos

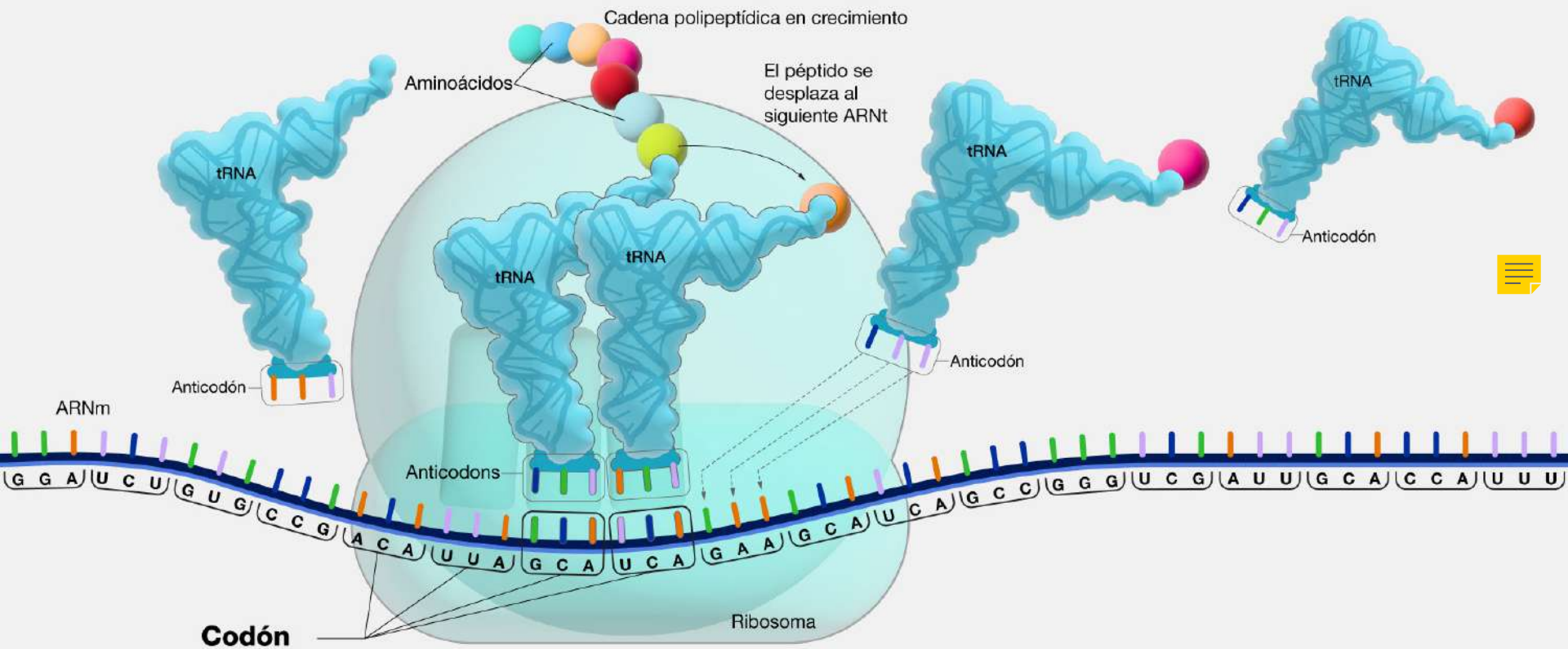


<https://youtu.be/bu0FXyRNptg>

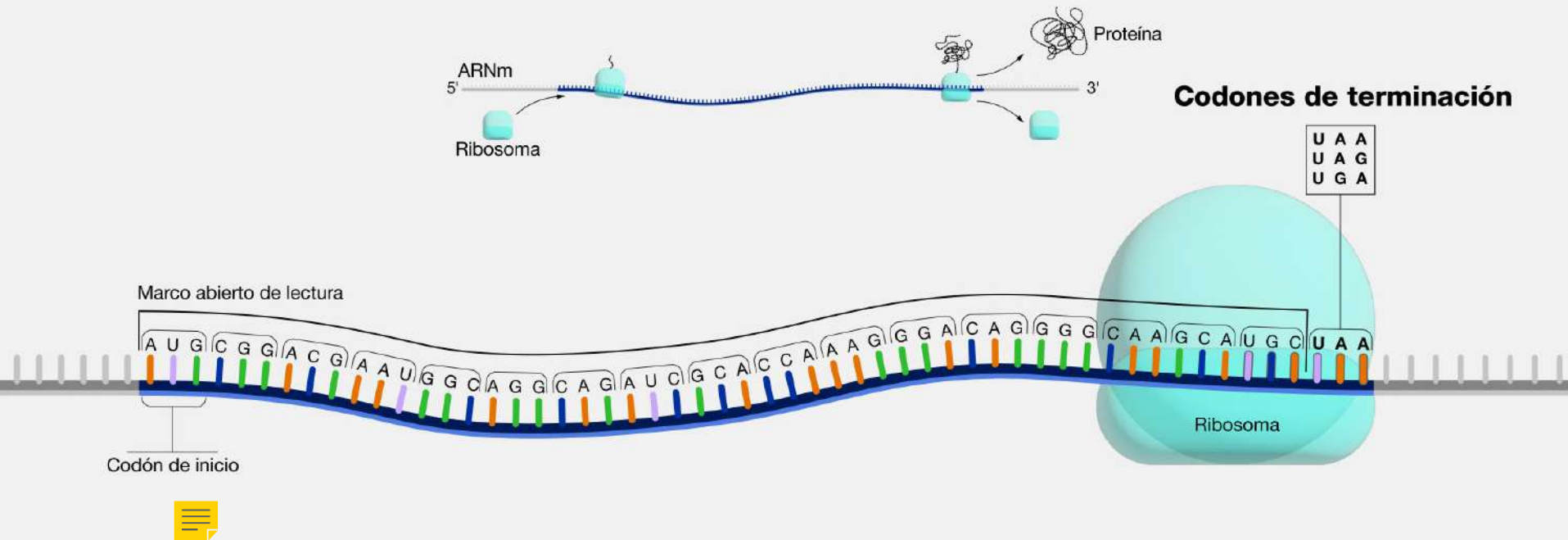
Cadena polipeptídica



Codón



Codón de terminación



El código genético en Python

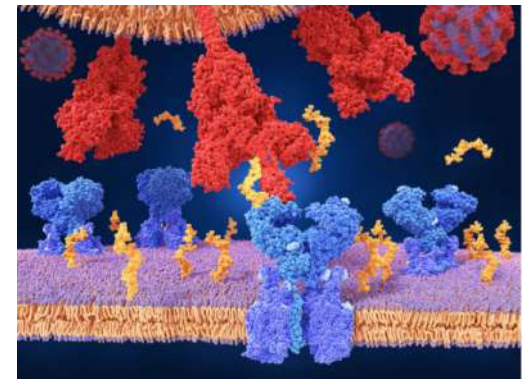
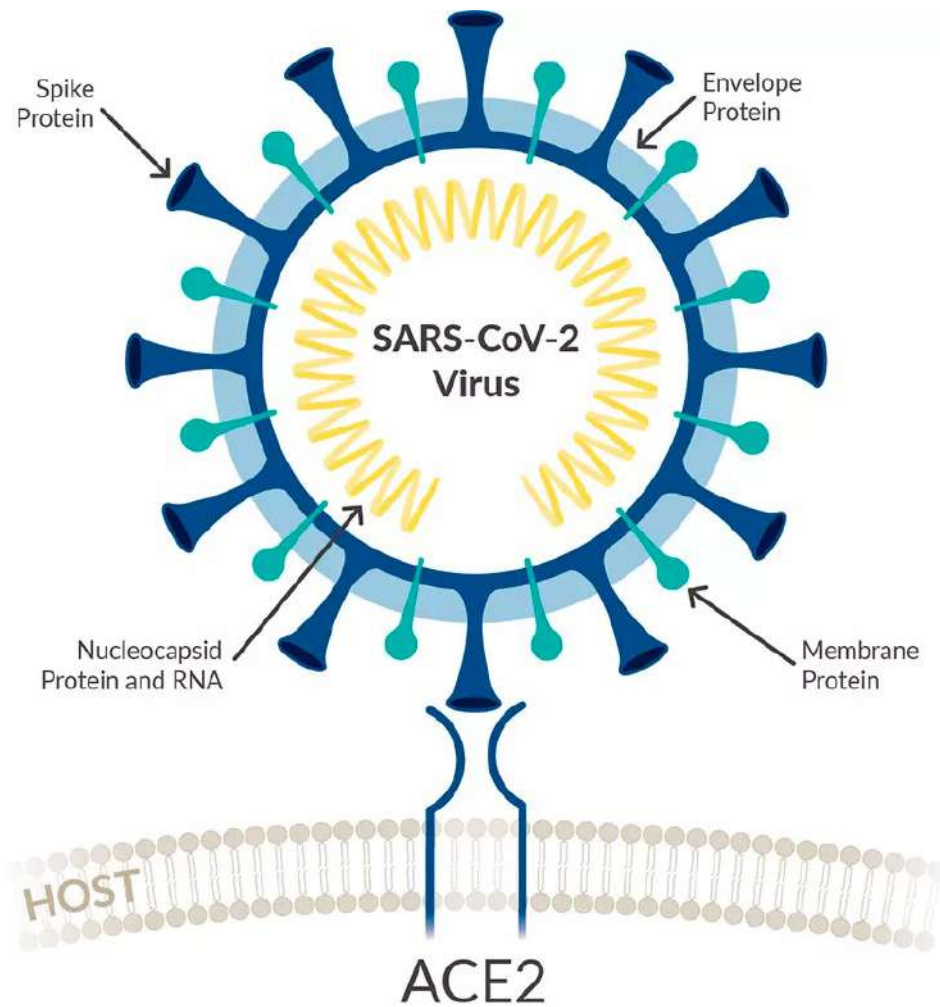
```
gencode = {
    'ATA': 'I', 'ATC': 'I', 'ATT': 'I', 'ATG': 'M',
    'ACA': 'T', 'ACC': 'T', 'ACG': 'T', 'ACT': 'T',
    'AAC': 'N', 'AAT': 'N', 'AAA': 'K', 'AAG': 'K',
    'AGC': 'S', 'AGT': 'S', 'AGA': 'R', 'AGG': 'R',
    'CTA': 'L', 'CTC': 'L', 'CTG': 'L', 'CTT': 'L',
    'CCA': 'P', 'CCC': 'P', 'CCG': 'P', 'CCT': 'P',
    'CAC': 'H', 'CAT': 'H', 'CAA': 'Q', 'CAG': 'Q',
    'CGA': 'R', 'CGC': 'R', 'CGG': 'R', 'CGT': 'R',
    'GTA': 'V', 'GTC': 'V', 'GTG': 'V', 'GTT': 'V',
    'GCA': 'A', 'GCC': 'A', 'GCG': 'A', 'GCT': 'A',
    'GAC': 'D', 'GAT': 'D', 'GAA': 'E', 'GAG': 'E',
    'GGA': 'G', 'GGC': 'G', 'GGG': 'G', 'GGT': 'G',
    'TCA': 'S', 'TCC': 'S', 'TCG': 'S', 'TCT': 'S',
    'TTC': 'F', 'TTT': 'F', 'TTA': 'L', 'TTG': 'L',
    'TAC': 'Y', 'TAT': 'Y', 'TAA': '-', 'TAG': '-',
    'TGC': 'C', 'TGT': 'C', 'TGA': '-', 'TGG': 'W'}

dna = 'CATGAAGTGTGTTGGTGGTTATTG'
long_dna = len(dna)
print('El DNA es: %s' % dna)
print('Longitud del DNA: %d' % long_dna)

proteina = ''
for i in range(0, long_dna, 3):
    codon = dna[i:i+3]
    proteina += gencode[codon]
print('La proteína es: %s' % proteina)
```

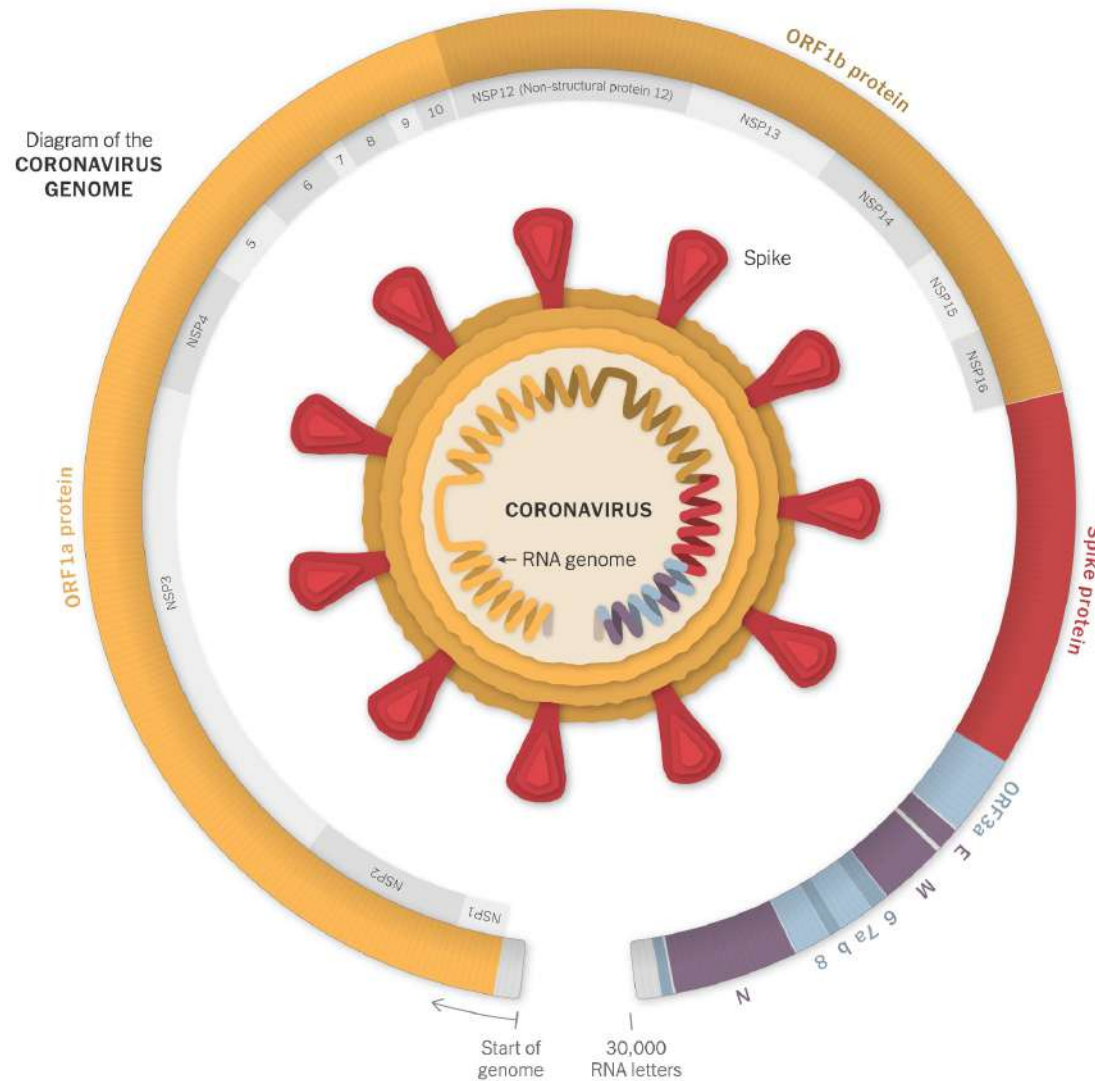
El DNA es: CATGAAGTGTGTTGGTGGTTATTG
Longitud del DNA: 24
La proteína es: HEVCWLL

Proteínas: ACE2 humano, Spike protein SARS-CoV-2



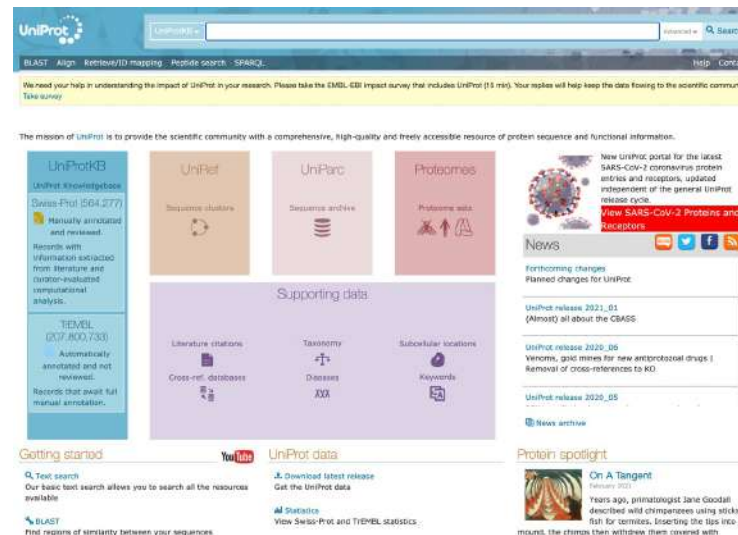
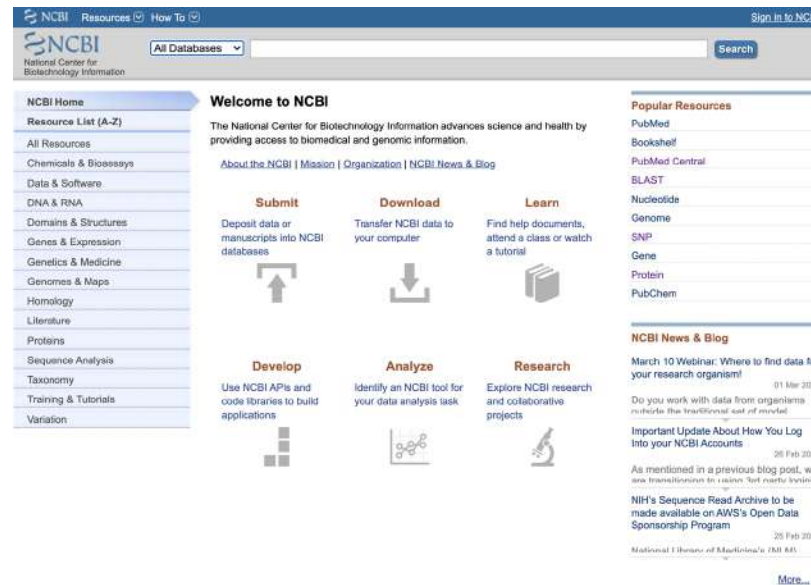
¿Cómo puedo obtener la secuencia de estas dos proteínas?

Spike protein



Práctica: descargar un fichero FASTA

- NCBI
- Uniprot



Práctica: leer un fichero FASTA

```
from sys import argv

def fasta2sec (fastafile) :
    """ Lee un fichero FASTA y
    devuelve la secuencia que contiene """
    F = open (fastafile)
    secuencia = ''
    for linea in F :
        if linea[0] != '>' :
            secuencia = secuencia + linea.strip('\n')
    return secuencia

sec = fasta2sec(argv[1])
lsec = len(sec)
print ('La secuencia tiene %d caracteres' % (lsec))

c4line = 80
for i in range (0, len(sec), c4line):
    print (sec[i:i+c4line])
```

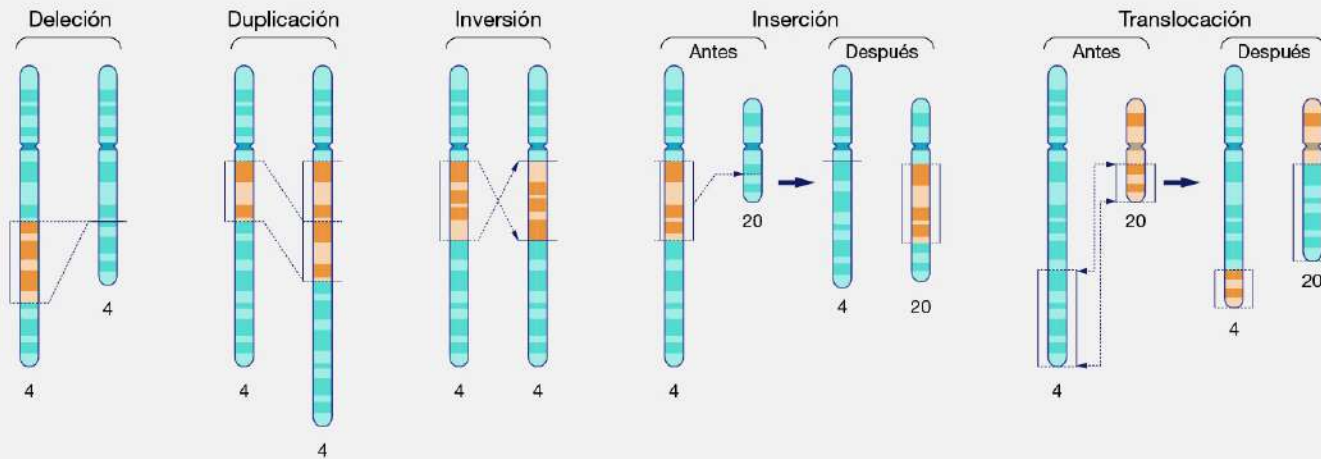
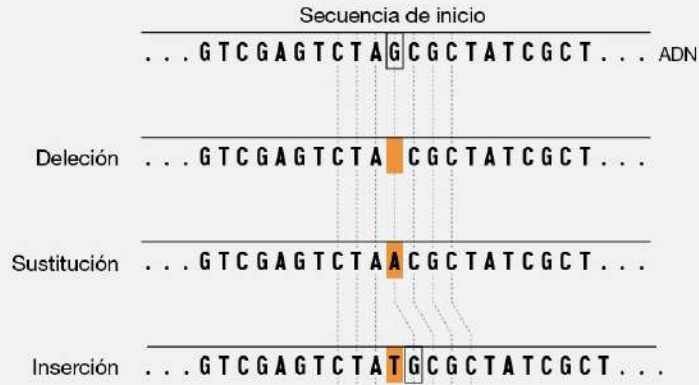
Práctica: Navegando por el genoma

- Servidores
 - UCSC Genome Browser
 - <https://genome.ucsc.edu/>
 - NCBI Genome Data Viewer
 - <https://www.ncbi.nlm.nih.gov/genome/gdv/>
 - EMBL-EBI Ensembl
 - <https://www.ensembl.org>
- Instalación local
 - Integrative Genomics Viewer (IGV)
 - <http://software.broadinstitute.org/software/igv/>
 - <https://igv.org/app>

Índice

- Células, genomas y cromosomas
- DNA
- Replicación
- Genes, transcripción y proteínas
- **Mutaciones**
- Genotipo y fenotipo

Mutaciones



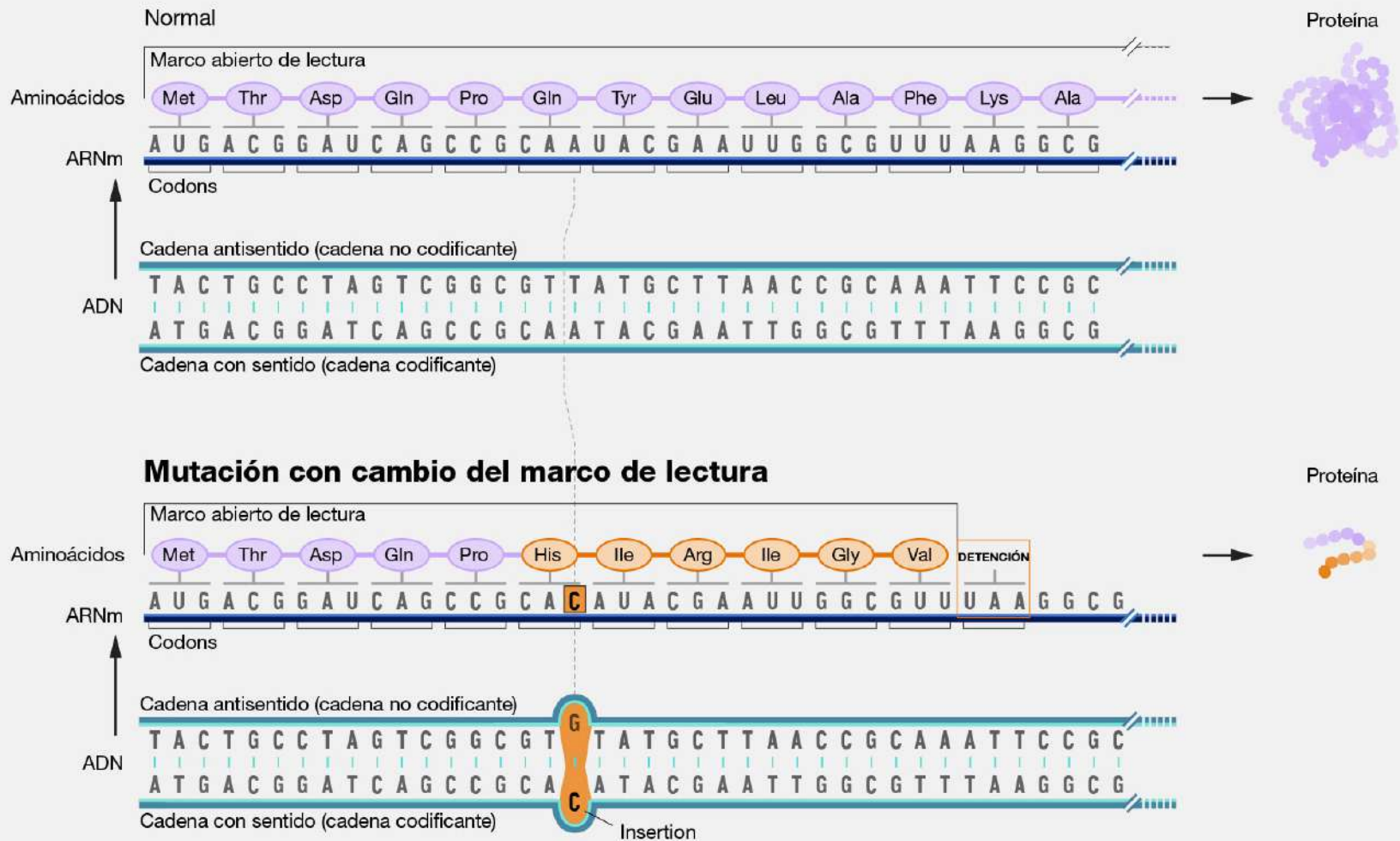
Mutaciones: *Missense* o cambio de sentido



Mutaciones: *Nonsense* o sin sentido



Mutaciones: *Frameshift* o cambio de marco



Práctica: Simulación de mutaciones con Python

```
import random

from sys import argv

def DNA2prot (sec) :
    gencode = {

    proteina = ''
    for i in range(0,len(sec),3) :
        codon = sec[i:i+3]
        proteina += gencode[codon]
    return proteina

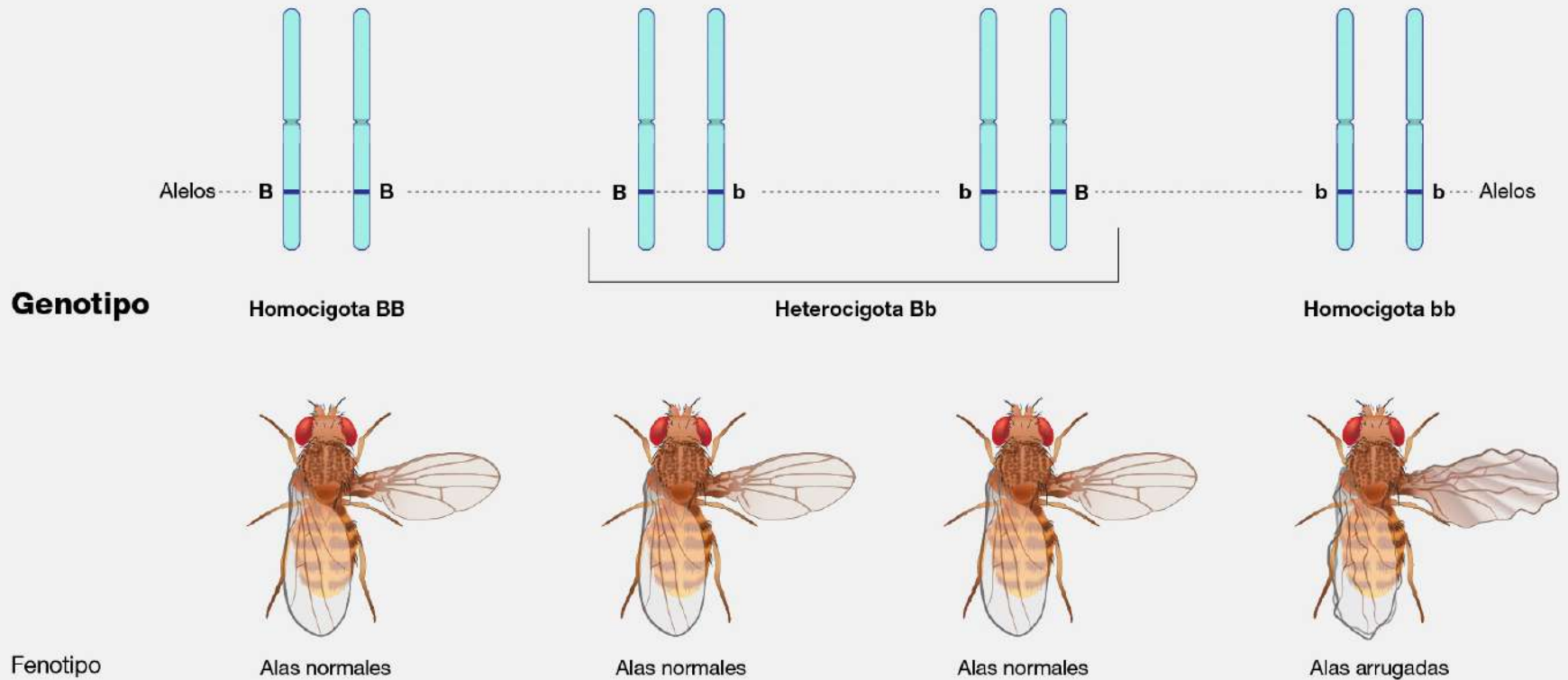
def nucaleatx (nuc) :
    nucs = ['A', 'T', 'C', 'G']
    nucs.remove(nuc)
    return random.choice(nucs)

DNA = argv[1]
print ('Proteina original: %s' % (DNA2prot(DNA)))
posmut = random.randrange(len(DNA))
new_nuc = nucaleatx(DNA[posmut])
DNA_mutado = DNA[:posmut] + new_nuc + DNA[posmut+1:]
print ('"Nueva" proteina: %s' % (DNA2prot(DNA_mutado)))
```

Índice

- Células, genomas y cromosomas
- DNA
- Replicación
- Genes, transcripción y proteínas
- Mutaciones
- Genotipo y fenotipo

Genotipo y fenotipo



Biología para informáticos

Máster en Sistemas Inteligentes

Luis Quintales
Departamento de Informática

